Faculty of Sciences of Sfax
Department of Mathematics

Sfax University

Laboratory of Probability
and Statistics





THESIS

Speciality: Applied Mathematics

Presented by

FAROUK MSELMI

Multivariate Normal α-Stable Distributions

SUPPORTED ON DECEMBER, 18TH, 2015 FRONT THE JURY

Mr. Mokhless HAMMAMI	PRESIDENT	Sfax University, Tunisia
Ms. Sonia FOURATI	REVIEWER	Rouen University, France
Mr. Mohamed Ali HAMMAMI	REVIEWER	Sfax University, Tunisia
Mr. Bassem BEN HAMED	EXAMINER	Sfax University, Tunisia
Mr. Afif MASMOUDI	SUPERVISOR OF THE THESIS	Sfax University, Tunisia
Mr. Mahdi LOUATI	CO-SUPERVISOR OF THE THESIS	Sfax University, Tunisia

Academic Year: 2014-2015

To my parents...

To my sisters....

To my brother...

To my friends...

Acknowledgements

It is not by tradition that this page figures in the preamble of this document, but it is rather moral duty and sincere recognition that push me to start with it. I would be indeed ungrateful if I don't express my recognition and my gratitude to all those who, from near or afar, have facilitated my task and permitted me to carry this work through.

I would especially like to express all my gratitude and my quick acknowledgments to my supervisor and my co-supervisor, Professor Afif MASMOUDI and Professor Mahdi LOUATI who granted me their confidence when proposing this topic of thesis and for their help with their experience and knowledge to make.

I would like also to express my sincere gratitude and thanks to Professor Mokhless HAMMAMI from Faculty of Sciences of Sfax, Tunisia, for accepting to be chairman of the exam committee of my thesis.

I extend my honorable thanks to Professor Mohamed Ali HAMMAMI, from Faculty of Science of Sfax, Tunisia, and Professor Sonia FOURATI from the National Institute of Applied Sciences of Rouen, France, for reporting this work and for their many valuable comments which helped improve this document.

For the honor they have provided by accepting to participate in the jury of this thesis, the time they have devoted and their insightful questions. I thank Professor Bassem BEN HAMED from Sfax National School of Electronics and Telecommunications, Tunisia.

A big THANK YOU to my Parents, who have given me the opportunity to pursue my studies and taught me to overcome the difficult moments, to my nice sisters Zaara, Asma, Marwa and Ghada and to my brother Saber who support and encourage me throughout my entire life.

I would like to thank all those who I have living and have studying with during these years especially to Aws, Lassed, Issa, Zahreddine, Bilel, Nabil and Souhail.

I thank all members of the Laboratory of Probability and Statistics of Faculty of Science of Sfax, especially to Hatem, Aymen, Lamia, Mouna, Sahar, Lobna, Rahma, Emna, Abdelaziz, Taher and Atizez...

Contents

Dedication i

Acknowledgements iii

Introduction 1

1 Preliminaries 7

1.1 Natural exponential families . . . 7

1.2 Infinitely divisible property . . . 10

1.3 Stable distributions . . . 13

1.4 Lévy process . . . 17

1.5 Multivariate Laplace distribution . . . 18

2 Multivariate normal α-stable exponential families 21

2.1 Normal α-stable random vector . . . 22

2.2 Variance function . . . 23

2.3 Lévy measures . . . 30

2.4 Simulation and estimations . . . 34

2.4.1 Parameter's estimations . . . 35

2.4.1.1 The method of moments . . . 35

2.4.1.2 The regression method . . . 36

2.4.2 Simulation of the normal α-stable distribution . . . 37

2.4.3 Estimation of parameters using the method of moments . . . 40

2.4.4 Estimation of parameters using the tailed regression method . . . 46

2.4.5 Comparison 54

3 Characterizations of multivariate stable processes. 57

3.1 Characterization of stable processes on $\mathbb{R}^d$ 57

3.2 Some characteristic properties of a stochastic process on $\mathbb{R}^d$ 63

Conclusion and perspectives 71

Bibliography 73

List of Figures

2.1 Curves of the normal $\alpha-$stable density with different values of α and β. 38

2.2 Histograms of the normal $\alpha-$stable distribution with sample size $N = 1000$. . . 39

2.3 Histograms of the estimator $\widehat{\alpha}$ by the MM for $K = 1000$ samples. 43

2.4 Histograms of the estimator $\widehat{\beta}$ by the MM for $K = 1000$ samples. 43

2.5 Curves of the $MSE(\widehat{\alpha})$ and $MSE(\widehat{\beta})$ by the MM when $\alpha = 0.2$, $\beta = 1$. 44

2.6 Curves of the $MSE(\widehat{\alpha})$ and $MSE(\widehat{\beta})$ by the MM when $\alpha = 0.5$, $\beta = 1$. 44

2.7 Curves of the densities $f_{\alpha,\beta}$ and $\widehat{f}$ by the MM. 45

2.8 Curves of the $MSE(\widehat{f})$ for $\alpha = 0.5$ and $\alpha = 0.7$. 46

2.9 Histograms of the estimator $\widehat{\alpha}$ by the TRM for $K = 1000$ samples. 51

2.10 Histograms of the estimator $\widehat{\beta}$ by the TRM for $K = 1000$ samples. 51

2.11 Curves of the $MSE(\widehat{\alpha})$ and $MSE(\widehat{\beta})$ by the TRM when $\alpha = 0.2$, $\beta = 1$. 52

2.12 Curves of the $MSE(\widehat{\alpha})$ and $MSE(\widehat{\beta})$ by the TRM when $\alpha = 0.5$, $\beta = 1$. 52

2.13 Curves of the densities $f_{\alpha,\beta}$ and $\widehat{f}$ by the TRM. 53

2.14 Curves of the $MSE(\widehat{f})$ for $\alpha = 0.5$ and $\alpha = 0.7$ 54

List of Tables

1.1 Cubic NEFs. 9
1.2 Quadratic NEFs. 9

2.1 Statistical study of the estimators, their MSEs, as well as their CVs 42
2.2 Statistical study of the estimators, their MSEs, as well as their CVs 50
2.3 The variation coefficient for different values of α and β. 55

Introduction

The class of stable distributions was introduced by Paul Lévy in 1924. The importance of stable probability distributions comes from the fact that they are "attractors" for properly normed sums of independent and identically-distributed (i.i.d) random variables (see Feller (1971)). In probability theory, a distribution is said to be stable (or a random variable is said to be stable) if a linear combination of two independent copies of a random sample has the same distribution.

By the classical Central Limit Theorem, the property normed sum of a set of random variables (each one with a finite variance) will tend towards a normal distribution as the number of variables increases. Without the finite variance assumption, the limit may be a stable distribution. This property is called the Generalized Central Limit Theorem (see Nolan (2007), page 21). Because of these important probabilistic properties, many studies have been carried out and several comprehensive books have been published. At this level, we may refer to Araujo and Giné (1980), Christoph and Wolf (1992), and Feller (1971).

The stable distributions were the apperent departure from normality along with the demand for a self-similar model for financial data (i.e., the shape of the distribution for yearly asset price changes should resemble that of the constituent daily or monthly price changes). The stable distributions are frequently used in the analysis of critical behavior and financial data. They appear particularly in finance and economics. Here, we may refer to Fama (1965), McCulloch (1996), Embrechts et *al.* (1997), and Nolan (2007).

In terms of Natural Exponential Families (NEFs), which represent a very important class of distributions in probability and statistical theory, the stable distributions have generated a NEF and some characteristic properties have been established (see Brown (1986), and Louati (2013)). In fact, in the last decades, the natural exponential families have attracted a considerable attention of researchers and several works have been realized. Different classifications of the NEFs according to the form of their variance functions have been achieved. For instance, Morris (1982) has described the real quadratic natural exponential families those the NEFs such that the variance function is a polynomial with degree less than or equal to two. Moreover, Letac and Mora (1990) have described the real cubic natural exponential families those the NEFs such that the variance function constitutes a polynomial with degree less than or equal to three. The characterizations of the multivariate versions of the Morris class and of the Letac-Mora class have also been realized, and were respectively due to Casalis (1990) and Hassairi (1992). Another important fact concerning the real stable distributions is that they appear as a part of the so called Tweedie scale which represents a class of NEFs such that the variance function is a power of the mean. More specifically,

$$V_F(m) = Bm^{\beta}; B > 0 \text{ and } \beta \in \mathbb{R} \setminus (0,1),$$

(see Letac (1992) and Seshadri (1994), page 165). Recently, Hassairi and Louati (2009) have characterized the class of multivariate stable exponential families by an homogeneous property of their variance function. In fact, they have studied separately the case $\alpha \neq 1$ and the case $\alpha = 1$. For $\alpha \neq 1$, they have shown that a necessary and sufficient condition for a multivariate distribution having a Laplace transform to be α-stable is that its cumulant function must satisfy the Euler differential equation of α-homogeneous functions. They have also shown that a multivariate α-stable exponential family was characterized by the fact that its variance function is $\frac{\alpha-2}{\alpha-1}$-homogeneous. For $\alpha = 1$, the cumulant function of a 1-stable distribution is not homogeneous, but it verifies a differential equation which is slightly different from the Euler equation.

In terms of processes, the class of stable processes is considered as a class of Lévy processes with stable distributions which is a stochastic process with independent and stationary increments. It represents the motion of a point whose successive displacements are random and independent, and are statistically identical over the different time intervals having the same length, and it may be viewed as the continuous-time analog of a random walk. Examples of stable processes include the Wiener process, or Brownian motion, whose associated probability distribution is the normal one. This class of processes is used in financial applications such as stock exchange, variation of stock market, financial returns, etc. (see Feller (1971), McCulloch (1996), Nolan (2003, 2007), Samorodnitsky and Taqqu (1994), Press (1972) and Zolotarev (1986, 1995)). This class of processes plays a major role and has a heavy tailed distribution. It appears in stochastic modelling in applied sciences, particularly in financial mathematics, and also in the theoretical motivation for the study of their properties (see Fama (1965), Fama and Roll (1971), McCulloch (1986) and Forsberg and Bollerslev (2002)).

In 1997, Barndorff-Nielsen defined the normal inverse Gaussian distribution by taking a random vector $(Y_1, \overline{Y})$ on $(0, +\infty) \times \mathbb{R}^{d-1}$ such that Y_1 is inverse Gaussian distributed, and the conditional distribution of $\overline{Y}$, given $Y_1 = y_1 > 0$ is a centered normal random vector on $\mathbb{R}^{d-1}$ with a variance-covariance matrix $y_1 I_{d-1}$, where I_{d-1} denotes the identity matrix of order $d-1$. The distribution of $\overline{Y}$ is called normal inverse Gaussian distribution. Several researchers have used this distribution in order to determine a Lévy process, which is representable through subordination of Brownian motion by the inverse Gaussian process.

Motivated by the construction of the normal inverse Gaussian distribution, we introduce a class of distributions called normal α-stable distributions with $\alpha \in (0, 1)$. This class represents an extension of the inverse Gaussian one ($\alpha = \frac{1}{2}$) by taking the random vector $(Y_1, \overline{Y})$ on $(0, +\infty) \times \mathbb{R}^{d-1}$ with a distribution $\mu_{\alpha,\beta}$ such that Y_1 is drifted α-stable distributed with a Laplace transform $L_{\pi_{\alpha,\beta}}(s) = e^{1-(1-\beta s)^{\alpha}}$, and the conditional distribution of $\overline{Y}$, given $Y_1 = y_1 > 0$ is a centered normal random vector on $\mathbb{R}^{d-1}$ with a variance-covariance matrix $y_1 I_{d-1}$. Hence, the distribution $\overline{\mu}_{\alpha,\beta}$ of $\overline{Y}$ is called the normal α-stable distribution. The importance of the normal α-stable distribution can be explained in financial modeling, since in this case

the error is not fixed. Then, we suppose that the log returns prices follows a normal α-stable distribution. In fact, we assume the conditional distribution of the log returns prices $\overline{Y}$ given the volatility $Y_1 = y_1$ is centered normal $N(0, y_1)$ and that the volatility Y_1 is a hidden random variable with drifted α-stable distribution $\pi_{\alpha,\beta}$. It follows that the mixture distribution of the log returns prices $\overline{Y}$ is given by

$$f_{\alpha,\beta}(\overline{y}) = \int_0^{+\infty} N(0, y_1)(\overline{y})\pi_{\alpha,\beta}(dy_1).$$

Since the importance of this distribution, we give some of its characteristic properties. So, we explicit the expression of the variance functions of the NEFs generated respectively by $\mu_{\alpha,\beta}$ and $\overline{\mu}_{\alpha,\beta}$. Since the distribution $\mu_{\alpha,\beta}$ of $(Y_1, \overline{Y})$ and the normal α-stable distribution of $\overline{Y}$ are infinitely divisible, then we may determine their associated Lévy measures.

Moreover, we propose an approximation of the normal α-stable distribution by using the Monte-Carlo method. As a result, we get the curves of the normal α-stable density and its cumulative function for different values of α and β. Furthermore, we obtain the histograms of this distribution by using the inverse transform sampling method. After studying the normal α-stable distribution, we estimate its parameters α and β by using the method of moments and the tailed regression one. We illustrate our estimations by their histograms for 1000 samples and we compute their mean squared error and their variation coefficient for each method. Hence, we get the curves of the mean squared error (MSE) of the estimators $\widehat{\alpha}$ and $\widehat{\beta}$, as well as the curves of the estimated and true densities for different values of α and β.

Recently, Letac and Seshadri (2005) have characterized the univariate drifted stable process $Y = (Y(t))_{t\geq 0}$ which is a Lévy process such that $Y(1)$ is drifted stable distributed. In fact, let $Y = (Y(t))_{t\geq 0}$ be a Lévy process on $(0, +\infty)$ and let T be a standard exponential random variable (*i.e.,* with density function $f_T(t) = e^{-t}1_{(0,+\infty)}(t)$) independent of Y, then for any fixed $p > 1$, $Y(T)/T^p$ and $Y(T)$ are independent if, and only if, $(Y(t))_{t\geq 0}$ is a drifted stable process.

So, using this result, we give a characterization of stable processes by using an independence property between $Y(T)$ and T, where $Y = (Y(t))_{t\geq 0}$ is a Lévy process and T is an

exponential random variable independent of Y. By referring to the above characterization of stable processes, we introduce two Lévy processes $(Y_1(t))_{t\geq 0}$ on $(0,+\infty)$ and $(\overline{Y}(t))_{t\geq 0}$ on $\mathbb{R}^{d-1}$, such that the first one follows a drifted stable distribution and, under the cut property due to Barndorff-Nielsen and Koudou (1995) and some independence conditions, we prove that $\overline{Y}(t)$, conditioned on $Y_1(t) = y_1 > 0$, follows a centered Gaussian distribution with a variance ty_1. Hence, it is important to notice that $(\overline{Y}(t))_{t\geq 0}$ follows a normal inverse Gaussian process. From this, we give some characteristic properties of the processes $(Y_1(t))_{t\geq 0}$ and $(\overline{Y}(t))_{t\geq 0}$ and we establish the link between the mixture distribution $\overline{Y}(T)$ and the Laplace one. Thesis is organized as follows.

In Chapter 1, we introduce some notations and we recall some facts concerning the natural exponential families and the cut property. Furthermore, we recall the characterization of the natural exponential families by their variance functions. So we illustrate this characterizations by some examples. We also introduce the infinitely divisible and stable distributions, and we present some properties of the stable distributions. We recall the definition of Lévy processes and we give the link between this class and the natural exponential families.

In Chapter 2, we use the normal inverse Gaussian distributions in order to introduce the class of multivariate normal α-stable distributions, and to establish some of their fundamental properties. We give the expression of the variance function of the generated natural exponential family and we use the Lévy-Khintchine representation which allows us to determine the associated Lévy measure. We also study the relationship between these distributions and the multivariate inverse Gaussian ones.Moreover, we approximate the normal α-stable distribution by the Monte-Carlo method. We get the curves of the normal α-stable density for different values of α and β. After studying the normal α-stable distribution, we estimate its parameters α and β by using both the method of moment and the tailed regression one. Then, we illustrate our estimations by their histograms for 1000 samples and we compute their mean squared error (MSE) and their variation coefficient for each method. Hence, we get the curves of the MSE of the estimators $\widehat{\alpha}$ and $\widehat{\beta}$, as well as the curves of the estimated and true densities for different values of (α, β).

Finally, in Chapter 3, we extend some characteristic properties of the stochastic Lévy processes $(Y(t))_{t\geq 0}$. Under some independence conditions between a standard exponential random variable T and $Y(T)$, we characterize the class of stable processes. By combining these results with the one given by Letac and Seshadri (2005), we introduce a class of multivariate stochastic processes and, we give the link between the distribution of $\overline{Y}(T)$ and the Laplace one. Moreover, we give some properties of the distribution $\overline{Y}(t)$ which is defined as a normal variance-mean mixture with a stable mixing distribution and we determine the distribution of $\widetilde{Y}(T)$. These results extend the one dimensional case given by Letac and Seshadri (2005).

Chapter 1

Preliminaries

In this chapter, we first recall some definitions and notations dealing with the natural exponential families (NEFs) and their variance functions. For more details, we may refer to Letac (1992). Then, we introduce the infinitely divisible and stable distributions. Some properties of these distributions are given.

1.1 Natural exponential families

A Natural Exponential Family (NEF) is a class of probability distributions which represents an exponentially titled version of a positive distribution. In fact, in the last decades, the natural exponential families have attracted a considerable attention of researchers and several works have been realized on their different aspects. As a result, different classifications of the NEFs according to the form of their variance functions have been achieved (see, for instance, Letac (1992), Letac and Mora (1990)).

Let μ be a positive measure on $\mathbb{R}^d$. We denote by

$$L_\mu(\theta) = \int_{\mathbb{R}^d} e^{\langle \theta, x \rangle} \mu(dx)$$

its Laplace transform, where $\langle \theta, x \rangle$ is the ordinary scalar product on $\mathbb{R}^d$ and $\Theta(\mu)$ is the interior of the convex set

$$D(\mu) = \{\theta \in \mathbb{R}^d;\ L_\mu(\theta) < +\infty\}.$$

$\mathcal{M}(\mathbb{R}^d)$ will denote the set of positive measures μ, such that μ is not concentrated on an affine hyperplane of $\mathbb{R}^d$ and $\Theta(\mu)$ is not empty.
For each $\mu \in \mathcal{M}(\mathbb{R}^d)$ and $\theta \in \Theta(\mu)$, we define the cumulant function of μ by

$$k_\mu(\theta) = \ln\left(L_\mu(\theta)\right).$$

The probability set

$$F = F(\mu) = \{P(\theta,\mu) = e^{\langle\theta,x\rangle - k_\mu(\theta)}\mu;\ \theta \in \Theta(\mu)\}$$

is called the NEF generated by μ. We also say that μ is a basis of F.
It is well known that the mean of $P(\theta,\mu)$ is

$$k'_\mu(\theta) = \int_{\mathbb{R}^d} xP(\theta,\mu)(dx). \tag{1.1}$$

The first derivative k'_μ defines a diffeomorphism between $\Theta(\mu)$ and its image $M_{F(\mu)}$, called the domain of the means of $F(\mu)$. We denote by

$$\psi_\mu :\ M_{F(\mu)} \longrightarrow \Theta(\mu)$$

its inverse function. The covariance operator $k''_\mu(\theta)$ of $P(\theta,\mu)$ is symmetric and definite positive. It is given by

$$k''_\mu(\theta) = \int_{\mathbb{R}^d} x \otimes xP(\theta,\mu)(dx) - k'_\mu(\theta) \otimes k'_\mu(\theta),$$

where, for all vectors u and v in $\mathbb{R}^d$, $(x \otimes x)(u,v) = \langle x,u\rangle\langle x,v\rangle$.
The map defined on $M_{F(\mu)}$ by $m \longmapsto V_{F(\mu)}(m) = k''_\mu(\psi_\mu(m))$ is called the variance function of the NEF $F(\mu)$. We may easily prove that, for all $m \in M_F$, we have

$$V_F(m) = (\psi'_\mu(m))^{-1}.$$

The importance of the variance function can be explained as follows: on the one hand, it is a function of the mean, and on the other hand, it characterizes the family F within the class of all natural exponential families (see Tweedie (1947)). Furthermore, for many common NEFs, the variance function takes a very simple form (see Kokonendji and Masmoudi (2006, 2013),

Hassairi (1992), Hassairi and Louati (2009), Hassairi (1999), Hassairi and Zarai (2004), Letac and Mora (1990), and Morris (1982)). An important feature of V_F is that it characterizes F in the following sense: If F_1 and F_2 are two NEFs such that $V_{F_1}(m)$ and $V_{F_2}(m)$ coincide on a nonempty open set of $M_{F_1} \cap M_{F_2}$, then $F_1 = F_2$. In particular, the knowledge of the variance function informs us about the NEF.

Next, we give some examples of the quadratic and cubic variance functions.

Example 1.1.1 *The following tabular represent the cubic and quadratic variance functions*

NEF	M_F	$V_F(m)$
Inverse Gaussian	$(0,+\infty)$	$\frac{m^3}{p^2}$
Ressel-Kendall	$(0,+\infty)$	$\frac{m^2}{p}(1+\frac{m}{p})$
Generalized Poisson	$(0,+\infty)$	$m(1+\frac{m}{p})^2$
Generalized negative Binomial	$(0,+\infty)$	$m(1+\frac{m}{p})(1+\frac{a+1}{a}\frac{m}{p})$
Strict Arcsinus	$(0,+\infty)$	$m(1+\frac{m^2}{p^2})$
Large Arcsinus	$(0,+\infty)$	$m(1+2\frac{m}{p}+\frac{a^2+1}{a^2}\frac{m^2}{p^2})$

Table 1.1: Cubic NEFs.

NEF	M_F	$V_F(m)$
Normal	$\mathbb{R}$	p
Poisson	$(0,+\infty)$	m
Gamma	$(0,+\infty)$	$\frac{m^2}{p}$
Binomial	$(0,N)$	$m(1-\frac{m}{N})$
Negative Binomial	$(0,+\infty)$	$m(1+\frac{m}{p})$
Hyperbolic cosinus	$\mathbb{R}$	$m(1+\frac{m^2}{p^2})$

Table 1.2: Quadratic NEFs.

Note that, Casalis (1990) has characterized the multivariate simple quadratic NEF by its variance function which has the form $am\otimes m+B(m)+c$, where B is linear on m and $a,c\in\mathbb{R}$ and he has shown that the Wishart variance function is the unique quadratic homogeneous variance function, that is $P(m)\theta = m\theta m$. Moreover, Hassairi and Zarai (2004) have characterized the cubic multivariate NEF using the orthogonality of polynomials.

Now, we focus on the notion of the cut in NEF introduced by Barndorff-Nielsen and Koudou (1995). Let $\mathcal{A}$ be the Borelian σ-algebra of $\mathbb{R}^d$ with $d \geq 2$, $\mathcal{P}$ the set of $\mathbb{R}^d$-valued probability measures and let $\mathcal{T}$ be a measurable map from $\mathbb{R}^d$ into $\mathbb{R}^k$ (with $1 \leq k < d$). We denote by $\mathbf{P}_{\mathcal{T}}$ the restriction of $\mathbf{P} \in \mathcal{P}$ on $\mathbb{R}^k$ and by $\mathbf{P}^{\mathcal{T}}$ the Markov kernel of the conditional distribution given $\mathcal{T}$ under $\mathbf{P} \in \mathcal{P}$. The map $\mathcal{T}$ is said to be a cut, if the mapping

$$\mathcal{C}_{\mathcal{T}} : \ \mathcal{P} \longrightarrow \{\mathbf{P}_{\mathcal{T}}; \mathbf{P} \in \mathcal{P}\} \times \{\mathbf{P}^{\mathcal{T}}; \mathbf{P} \in \mathcal{P}\}; \ \mathbf{P} \longmapsto (\mathbf{P}_{\mathcal{T}}, \mathbf{P}^{\mathcal{T}})$$

is surjective.

Let $\mu(dy_1, d\overline{y})$ be the probability distribution of $(Y_1(1), \overline{Y}(1))$ on $(0, +\infty) \times \mathbb{R}^{d-1}$ and let us consider $p_1 : \mathbb{R} \times \mathbb{R}^{d-1} \longrightarrow \mathbb{R}$; $(y_1, \overline{y}) \longmapsto y_1$ as a canonical projection on $\mathbb{R}$. The NEF $F = F(\mu) = \{\nu(dx) = e^{\langle \theta, x \rangle - k_\mu(\theta)} \mu(dx); \theta \in \Theta(\mu)\}$ has a cut on $\mathbb{R}$ if, and only if, $p_1(F) = \{p_1(\nu); \nu \in F\}$ is also a NEF on $\mathbb{R}$. For all disintegrations,

$$\mu(dy_1, d\overline{y}) = \pi(dy_1) Q_{y_1}(d\overline{y})$$

of μ, where Q_{y_1} denotes the conditional distribution of $\overline{Y}(1)$ given $Y_1(1) = y_1$, and π is the marginal distribution of $Y_1(1)$, Barndorff-Nielsen and Koudou (1995) (Theorem 3.1 (1)$\Leftrightarrow$(5)$\Leftrightarrow$(6)) show that $F = F(\mu)$ has a cut on $\mathbb{R}$ if, and only if, there exist $H : \Theta(Q_{y_1}) \subset \mathbb{R}^{d-1} \longrightarrow \mathbb{R}$ and $K : \Theta(Q_{y_1}) \subset \mathbb{R}^{d-1} \longrightarrow \mathbb{R}$, such that the conditional cumulant function of the distribution Q_{y_1} has the form

$$k_{Q_{y_1}}(\overline{\theta}) = K(\overline{\theta}) - y_1 H(\overline{\theta}), \ \overline{\theta} \in \Theta(Q_{y_1}). \tag{1.2}$$

1.2 Infinitely divisible property

In probability theory, a probability distribution is called infinitely divisible if it can be expressed as the probability distribution of the sum of an arbitrary number of independent and identically distributed random variables (see Sato (1999)). More precisely, we have

Definition 1.2.1 *Let μ be a probability measure on $\mathbb{R}^d$. μ is said to be infinitely divisible if, for all $n \in \mathbb{N}^*$, there exists a probability measure μ_n such that*

$$\mu_n^{*n} = \underbrace{\mu_n * \mu_n * \ldots * \mu_n}_{n \text{ times}} = \mu.$$

The measure μ is the nth power of convolution of the measure μ_n.

In terms of random variables, we have the following definition.

Definition 1.2.2 *A random variable X has an infinitely divisible distribution if, and only if, for all $n \in \mathbb{N}^*$, there exist $X_1, \ldots, X_n$ i.i.d such that*

$$X_1 + \ldots + X_n =^{\mathcal{L}} X,$$

where $=^{\mathcal{L}}$ represents the equality in distribution.

The following theorem characterizes the class of infinitely divisible distributions and it was given by Shiryayev (1984), page 336.

Theorem 1.2.3 *A random variable X is the limit of the sum of n random variables i.i.d. if, and only if, X is infinitely divisible.*

Let us notice that, the Laplace transform of an infinitely divisible distribution can be written as an nth power of another Laplace transform. In fact, if μ is an infinitely divisible distribution, then for all $n \in \mathbb{N}^*$, there exists a distribution μ_n such that

$$L_\mu(\theta) = (L_{\mu_n}(\theta))^n, \text{ for all } \theta \in \Theta(\mu).$$

The very famous examples of infinitely divisible distributions are:

Examples 1.2.4

1. *The centered normal distribution $N(0, \sigma^2)$: The Laplace transform is given by*

$$L_{N(0,\sigma^2)}(t) = \exp\left(\frac{t^2\sigma^2}{2}\right) = \left[\exp\left(\frac{t^2\frac{\sigma^2}{n}}{2}\right)\right]^n$$

   and it represents the nth power of the Laplace transform of the centered normal distribution $N(0, \frac{\sigma^2}{n})$.

2. *The Poisson distribution $\mathcal{P}(\lambda)$:*

$$L_{\mathcal{P}(\lambda)}(t) = \exp\left(\lambda(e^t - 1)\right) = \left[\exp\left(\frac{\lambda}{n}(e^t - 1)\right)\right]^n.$$

   It represents the Laplace transform of the nth power of convolution of the Poisson $\mathcal{P}(\frac{\lambda}{n})$.

3. *The Gamma distribution $\gamma(r,\lambda)$: The Laplace transform is given by*

$$L_{\gamma(r,\lambda)}(t) = \frac{1}{\left(1-\frac{t}{\lambda}\right)^r} = \frac{1}{\left[\left(1-\frac{t}{\lambda}\right)^{\frac{r}{n}}\right]^n}$$

and it represents the nth power of the Laplace transform of the Gamma $\gamma(\frac{r}{n},\lambda)$.

Let μ be a positive measure on $\mathbb{R}^d$. The set

$$\Lambda(\mu) = \{\lambda > 0;\ \exists\ \mu_\lambda \text{ such that } k_{\mu_\lambda}(\theta) = \lambda k_\mu(\theta), \forall \theta \in \Theta(\mu_\lambda) = \Theta(\mu)\}$$

is called the Jørgensen set or the Wallach set of μ. The measure μ is infinitely divisible if, and only if, the set $\Lambda(\mu)$ is equal to $(0,+\infty)$ (see Seshadri (1994), page 155).
Next, we recall the classification theorem of infinitely divisible measures in $\mathcal{M}(\mathbb{R}^d)$. That is the Lévy-Khintchine representation

Theorem 1.2.5 *Let $\mu \in \mathcal{M}(\mathbb{R}^d)$. Then μ is infinitely divisible if, and only if, there exist a symmetric positive matrix Σ and a measure ν satisfies the condition*

$$\int_{\mathbb{R}^d\setminus\{0\}} \min(1,\|x\|^2)\nu(dx) < +\infty$$

such that

$$k''_\mu(\theta) = \Sigma + \int_{\mathbb{R}^d\setminus\{0\}} x\otimes x e^{\langle\theta,x\rangle}\nu(dx).$$

The measure ν is called the Lévy measure.

Next, we give some examples of Lévy measures

Examples 1.2.6

1. *The normal distribution $N(m,\sigma^2)$: The parameter Σ is equal to σ^2 and the Lévy measure is 0.*

2. *The Poisson distribution $\mathcal{P}(\lambda)$: The parameter Σ is equal to 0 and the Lévy measure is $\lambda\delta_1(dx)$, where δ_1 is the Dirac measure on 1.*

3. *The Gamma distribution $\gamma(r,\lambda)$ with $r>0$ and $\lambda\in\mathbb{R}$: The parameter Σ is equal to 0 and the Lévy measure is $rx^{-1}e^{-\lambda x}1_{(0,+\infty)}(x)dx$.*

1.3 Stable distributions

The class of stable distributions introduced by Paul Lévy in 1924 in the course of his study of the sum of independent random variables. It is a rich class of probability distributions which has several mathematical properties. The stable distributions play a constantly increasing role. They are particularly important in probability because of their role as a natural generalization of the normal distribution. They are also important in the practical work of the statisticians in the analysis of data in several areas of applications (see Fama (1965), and Nolan (2007)).

Definition 1.3.1 *Let α be an element of $(0, 2]$. A random variable X on $\mathbb{R}$ is said to be α-stable if, for each $n \geq 2$, there exist $f_n \in \mathbb{R}$ and n independent copies $X_1, X_2, \ldots, X_n$ of X, such that*

$$X_1 + X_2 + \ldots + X_n =^{\mathcal{L}} n^{1/\alpha} X + f_n. \tag{1.3}$$

Moreover, X is strictly stable if (1.3) holds with $f_n = 0$.

Note that the case $\alpha = 2$ corresponds to the Gaussian random variable. It is the unique stable distribution with a finite variance. Let us consider $\alpha \in (0, 1)$. A positive random variable X has a standard positive stable distribution with index α, if there exists $c > 0$ such that, for all $\theta_1 \leq 0$, we have

$$L_X(\theta_1) = \mathbb{E}(e^{\theta_1 X}) = e^{-c(-\theta_1)^\alpha}.$$

(For more details, see Feller (1971), page 448).

Examples 1.3.2

1. *The standard normal distribution with a density*

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

   is stable with an index $\alpha = 2$.

2. *The Cauchy distribution with scale parameter $t > 0$ has the density*

$$f_t(x) = \frac{1}{\pi} \frac{t}{t^2 + (x - m)^2},$$

   where $m \in \mathbb{R}$. The Cauchy distribution is stable with an index $\alpha = 1$ (see Feller (1971), page 173).

3. The inverse Gaussian distribution with density

$$f(x) = \frac{x^{-\frac{3}{2}}}{\sqrt{2\pi}} e^{-\frac{1}{2x}} 1_{(0,+\infty)}(x)$$

is stable with an index $\alpha = \frac{1}{2}$. *(see Feller (1971), page 173 or Seshadri (1994), page 39).*

Remark 1.3.3 *The case where the index* α *tends to* 0 *corresponds to the Gamma distribution.*

As example of NEF generated by the stable distribution, we have the inverse Gaussian one. In fact, let μ be the distribution of an inverse Gaussian random variable. then, we have

1. $k_\mu(\theta) = -\sqrt{-2\theta},\ \theta \leq 0.$
2. $k'_\mu(\theta) = m = (-2\theta)^{-1/2},\ M_{F(\mu)} = (0, +\infty).$
3. $\psi_\mu(m) = -(2m^2)^{-1}$ and $V_{F(\mu)}(m) = m^3.$

The NEF generated by μ is

$$F = \{P(\theta, \mu);\ \theta \in \Theta(\mu)\}$$

where

$$P(\theta, \mu)(dx) = \frac{x^{-\frac{3}{2}}}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2x} + \theta x + \sqrt{-2\theta}\right\} 1_{(0,+\infty)}(x)dx$$

and we have the following parametrization

$$P(m, F)(dx) = \frac{x^{-\frac{3}{2}}}{\sqrt{2\pi}} \exp\left\{-\frac{(x-m)^2}{2m^2 x}\right\} 1_{(0,+\infty)}(x)dx.$$

Next, we give the link between a stable distribution and an infinitely divisible one.

Proposition 1.3.4 *If* X *is stable, then* X *is infinitely divisible.*

It is important to notice that the stable distribution is characterized by its Laplace transform. In fact, if μ is a real stable distribution, then its Laplace transform, when it exists, is given by

$$L_\mu(\theta) = \begin{cases} \exp\left(\frac{c\Gamma(3-\alpha)}{\alpha(\alpha-1)}|\theta|^\alpha\right), & \alpha \in (0, 2] \setminus \{1\} \\ \exp\left(c|\theta| \ln|\theta| + b|\theta|\right), & \alpha = 1 \end{cases}$$

Remark 1.3.5 *The inverse of Proposition 1.3.4 is false. The Poisson distribution is infinitely divisible but it is not stable.*

We say that a random variable Z has a drifted positive stable distribution (see Hougaard (1986)), if there exists $\theta_0 < 0$ such that, for all $\theta_1 \leq -\theta_0$, we have

$$L_Z(\theta_1) = \mathbb{E}(e^{\theta_1 Z}) = e^{-c(-\theta_1-\theta_0)^\alpha + c(-\theta_0)^\alpha}. \tag{1.4}$$

This represents an exponentially titled version of the stable distribution and it called also tempered stable distribution (see Barndorff-Nielsen and Shephard (2001)).

Remarks 1.3.6
1. Note that the drifted distribution belongs to the exponential family generated by the standard one. In fact, according to (1.4), we can write, for all $\theta_1 \leq -\theta_0$,

$$L_Z(\theta_1) = e^{c(-\theta_0)^\alpha} e^{-c(-\theta_1-\theta_0)^\alpha} = e^{c(-\theta_0)^\alpha} L_X(\theta_1+\theta_0).$$

This implies that the distribution P_Z of Z is an element of the NEF generated by P_X:

$$P_Z(dz) = e^{\theta_0 z + c(-\theta_0)^\alpha} P_X(dz).$$

2. The Laplace transform of a drifted positive stable random variable Z has the following form

$$\begin{aligned} L_Z(\theta_1) &= e^{-c(-\theta_1-\theta_0)^\alpha + c(-\theta_0)^\alpha} \\ &= e^{c(-\theta_0)^\alpha (1-(1+\frac{\theta_1}{\theta_0})^\alpha)} \\ &= e^{t(1-(1-\beta\theta_1)^\alpha)}, \text{ for all } \theta_1 < \frac{1}{\beta}, \end{aligned}$$

where $\beta = -\frac{1}{\theta_0} > 0$ and $t = c(-\theta_0)^\alpha > 0$. Note that, without loss of generality, we can take $t = 1$ (see Letac and Seshadri (2005)).

The class of multivariate stable random variables is defined as an extension to $\mathbb{R}^d$ or to any Euclidean space of the univariate one.

Definition 1.3.7 *A random vector X is stable if, for any $n \geq 2$, there is a vector $f_n \in \mathbb{R}^d$ such that*

$$X_1 + ... + X_n =^{\mathcal{L}} n^{\frac{1}{\alpha}} X + f_n,$$

where $X_1, ..., X_n$ are independent copies of X and $\alpha \in (0, 2]$, called the index of stability of the distribution (see Zolotarev (1986)).

The stable distributions are also characterized by their Laplace transforms when they exist, parameterized by (α, ρ, δ), where α is the index, δ in $\mathbb{R}^d$ is called the shift vector, and ρ represents a finite measure concentrated on the unit sphere $\mathbb{S}$ of $\mathbb{R}^d$ called the spectral measure.

For a subset A of $\mathbb{R}^d$, we denotes by $CCH(A)$ the closed convex hull of A and by $Cone(A)$ the cone generated by A, that is

$$Cone(A) = \{x \in \mathbb{R}^d;\ x = ra \text{ for some } r > 0 \text{ and } a \in A\}.$$

Then, the support of a stable distribution with a spectral measure ρ depends on the position of the index parameter $\alpha \in (0, 2]$. In fact, when $\alpha < 1$, the distribution is concentrated on the cone $(Cone(CCH(support(\rho)))) + \delta$, and when $\alpha \geq 1$, the distribution is concentrated on the whole space. It is shown (see Nolan (2007), page 197) that if the support of the spectral measure ρ is contained in some half-space, then the Laplace transform of the stable distribution exists and is given explicitly by the following proposition.

Proposition 1.3.8 *Let $\alpha \in (0, 2]$ and let μ be an α-stable distribution. Then, for all $\theta \in \Theta(\mu) = -CCH(Cone(support(\rho)))$,*

$$L_\mu(\theta) = \begin{cases} \exp\left(-\sec\left(\frac{\pi\alpha}{2}\right)\gamma(\theta)^\alpha\right) & if\ \alpha \neq 1 \\ \exp\left(\frac{2}{\pi}\int_{\mathbb{S}} \langle\theta, x\rangle \ln\left(|\langle\theta, x\rangle|\right)\rho(dx)\right) & if\ \alpha = 1, \end{cases}$$

where $\sec\left(\frac{\pi\alpha}{2}\right) = \left(\cos\left(\frac{\pi\alpha}{2}\right)\right)^{-1}$, $\gamma(\theta) = \left(\int_{\mathbb{S}} |\langle\theta, x\rangle|^\alpha \rho(dx)\right)^{\frac{1}{\alpha}}$.

In 2009, Hassairi and Louati have characterized the stable distribution using the Euler differential equation of α-homogenous functions and they have considered separately the case $\alpha \neq 1$ and $\alpha = 1$. Note that, a NEF is said to be α-stable if it is generated by an α-stable distribution.

Theorem 1.3.9 *Let F be a natural exponential family on $\mathbb{R}^d$, and $\alpha \in (0, 2]\backslash\{1\}$. Then the following properties are equivalent*

i) F is α-stable.

ii) F has a basis μ such that the cumulant function k_μ is $\alpha-$homogeneous.

iii) F has a basis μ such that for all $\theta \in \Theta(\mu)$, $\langle k'_\mu(\theta), \theta\rangle = \alpha k_\mu(\theta)$.

Note that the situation is slightly different when $\alpha = 1$. In fact, the cumulant function of a 1-stable distribution μ is not homogenous, however, it is characterized by a differential equation close to the Euler one.

Theorem 1.3.10 *Let F be a natural exponential family on $\mathbb{R}^d$. F is 1-stable if, and only if, it admits a basis μ such that*

$$\langle k'_\mu(\theta), \theta \rangle = k_\mu(\theta) + \langle \theta, m_0 \rangle, \tag{1.5}$$

where m_0 is a nonnul vector defined by

$$m_0 = \frac{2}{\pi} \int_{\mathbb{S}} s\rho(ds) \neq 0. \tag{1.6}$$

1.4 Lévy process

In this section we introduce the class of Lévy processes and we give the link between this class and the stable one.

Definition 1.4.1 *A stochastic process $(Y(t))_{t\geq 0}$ on $\mathbb{R}^d$ is a Lévy process, if the following conditions are satisfied:*

1. *$Y(0) = 0$ almost surely (a.s.).*
2. *For any choice of $n \geq 1$ and $0 < t_1 < \ldots < t_n$, the random variables $Y(t_1), Y(t_2) - Y(t_1), \ldots, Y(t_n) - Y(t_{n-1})$ are independent.*
3. *The distribution of $Y(t+s) - Y(s) =^{\mathcal{L}} Y(t)$.*
4. *For every $\omega \in \Omega$, $t \longmapsto Y(t, \omega)$ is càdlàg (i.e., right-continuous in $t \geq 0$ and has left-limits in $t > 0$.)*

In terms of natural exponential family, let us consider a Lévy process $(Y(t))_{t\geq 0}$ with distribution μ_t. Hence, we define the process

$$Z(t) = \exp\{\langle \theta, Y(t) \rangle - tk_{\mu_1}(\theta)\},\ \theta \in \Theta(\mu_1)$$

which has the distribution $P(\theta, \mu_t) \in F(\mu_t)$, where $F(\mu_t)$ represents the natural exponential family generated by μ_t. It is important to notice that the process $(Z(t))_{t\geq 0}$ is a martingale for every $\theta \in \Theta(\mu_1)$. As example, let $(B(t))_{t\geq 0}$ be a Brownian motion on $\mathbb{R}^d$ which is a

Lévy process with normal distribution $N(0, tI_d)$, so its cumulant function $k_{\mu_1}(\theta) = \frac{\|\theta\|^2}{2}$ and $\Theta(\mu_1) = \mathbb{R}^d$. Then we obtain the geometric Brownian motion

$$Z(t) = \exp\left\{\langle\theta, B(t)\rangle - t\frac{\|\theta\|^2}{2}\right\}.$$

Since $Y = (Y(t))_{t\geq 0}$ is a Lévy process with a distribution μ_t, then μ_t is infinitely divisible and consequently, it is characterized by its cumulant function, which is given by the Lévy-Khintchine formula:

$$k''_{\mu_t}(\theta) = t\left[\Sigma + \int_{\mathbb{R}^d\setminus\{0\}} x \otimes x e^{\langle\theta,x\rangle}\nu(dx)\right],$$

where Σ is a symmetric positive matrix and ν is the Lévy measure.

An interesting particular Lévy process is the stable one.

Definition 1.4.2 *Let $(Y(t))_{t\geq 0}$ be a process on $\mathbb{R}^d$. $(Y(t))_{t\geq 0}$ is an α-stable process, if it is a Lévy process such that $Y(1)$ is an α-stable random variable.*

Theorem 1.4.3 *Let $X(t)$ be an α-stable process with a distribution μ_t. Then, for all $t > 0$, the Laplace transform of $X(t)$ is given as follows, for all $\theta \in \Theta(\mu_1)$,*

$$L_{\mu_t}(\theta) = e^{tk_{\mu_1}(\theta)} = \begin{cases} \exp\left(-t\sec\left(\frac{\pi\alpha}{2}\right)\gamma(\theta)^\alpha\right) & if\ \alpha \neq 1 \\ \exp\left(t\frac{2}{\pi}\int_{\mathbb{S}}\langle\theta,x\rangle\ln\left(|\langle\theta,x\rangle|\right)\rho(dx)\right) & if\ \alpha = 1. \end{cases} \tag{1.7}$$

For more details, see Nolan (2007), page 197.

1.5 Multivariate Laplace distribution

In this part, we recall some properties concerning the multivariate Laplace distribution which is related to the modified Bessel function of the third Kind. Note that, the univariate Laplace distribution is a continuous probability distribution named after Pierre-Simon Laplace. It is also sometimes called the double exponential distribution (see Johnson *et al.* (1994), page 60). More precisely, A random variable has a Laplace distribution with parameters $(\mu, b) \in \mathbb{R} \times (0, +\infty)$, if its probability density function is

$$f(x \mid \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right) = \frac{1}{2b} \begin{cases} \exp\left(-\frac{\mu - x}{b}\right) & \text{if } x < \mu \\ \exp\left(-\frac{x-\mu}{b}\right) & \text{if } x \geq \mu \end{cases}.$$

Here, μ is a location parameter and $b > 0$, which is sometimes referred to as the diversity, is a scale parameter. If $\mu = 0$ and $b = 1$, the positive half-line is exactly an exponential distribution scaled by $1/2$.

Now, we may focus our attention on the multivariate Laplace distribution.

Definition 1.5.1 *Let $\lambda \in \mathbb{R}^*$, $a > 0$ and $b > 0$. The modified Bessel function of the third kind with order λ is given by*

$$K_\lambda(b) = \frac{1}{2} \int_0^{+\infty} (b/a)^\lambda \, x^{\lambda-1} e^{-\frac{1}{2}\left(\frac{a}{x} + \frac{b^2}{a} x\right)} dx. \tag{1.8}$$

(For more details about this function, the reader can see Seshadri (1994), page 27).

Definition 1.5.2 *Let σ be a vector of $\mathbb{R}^{d-1}$ and let Γ be a $(d-1)-$symmetric definite positive matrix, the multivariate Laplace distribution on $\mathbb{R}^{d-1}$, with a location parameter σ and a variance-covariance parameter Γ, is*

$$f_{(\sigma,\Gamma)}(y) = \frac{1}{(2\pi)^{\frac{d-1}{2}}} \frac{2K_{\frac{d-3}{2}}\left(\sqrt{2(y-\sigma)'\Gamma^{-1}(y-\sigma)}\right)}{\left(\sqrt{\frac{1}{2}(y-\sigma)'\Gamma^{-1}(y-\sigma)}\right)^{\frac{d-3}{2}}}.$$

This distribution is characterized by its Laplace transform equal to, for all $\theta \in \{\theta \in \mathbb{R}^{d-1} \; ; \; \theta'\Gamma\theta < 2\}$,

$$L_{\sigma,\Gamma}(\theta) = \frac{e^{\langle \sigma, \theta \rangle}}{1 - \frac{1}{2}\theta'\Gamma\theta}. \tag{1.9}$$

(see Torbjørn *et al.* (2006)).

Chapter 2

Multivariate normal α-stable exponential families

The normal inverse Gaussian distributions is an absolutely continuous probability distribution that is defined as the normal variance-mean mixture where the mixing density is the inverse Gaussian distribution. In terms of processes, this distribution defines a Lévy process, which is representable through subordination of Brownian motion by the inverse Gaussian process. Motivated by this construction, we introduce a class of distributions called normal α-stable distributions with $\alpha \in (0,1)$. This class represents the natural extension of the inverse Gaussian one ($\alpha = \frac{1}{2}$) by considering a random vector $(Y_1, \overline{Y})$ on $(0,+\infty)\times\mathbb{R}^{d-1}$ with a distribution $\mu_{\alpha,\beta}$ such that Y_1 is drifted α-stable distributed (its distribution is denoted by $\pi_{\alpha,\beta}$ and it is defined in Remark 1.3.6), and the conditional distribution of $\overline{Y}$, given $Y_1 = y_1 > 0$ is a centered normal random vector on $\mathbb{R}^{d-1}$ with a variance-covariance matrix $y_1 I_{d-1}$. The distribution $\overline{\mu}_{\alpha,\beta}$ of $\overline{Y}$ is called the normal α-stable distribution. We give the expression of the variance function of the generated natural exponential family and we use the Lévy-Khintchine representation to determine the associated Lévy measure. We also study the relationship between these distributions and the multivariate inverse Gaussian ones. In what follows, we denote by $(e_1, \ldots, e_d)$ an orthonormal basis of $\mathbb{R}^d$ and for each element x of $\mathbb{R}^d$ we use the decomposition $x = (x_1, \overline{x}) \in \mathbb{R}\times\mathbb{R}^{d-1}$.

2.1 Normal α-stable random vector

Let $F_{\alpha,\beta} = F(\mu_{\alpha,\beta})$ be the NEF generated by $\mu_{\alpha,\beta}$. Next, we characterize the distribution $\overline{\mu}_{\alpha,\beta}$ of the random vector $\overline{Y}$. More precisely, we have the following result.

Proposition 2.1.1 *The distribution $\overline{\mu}_{\alpha,\beta}$ of $\overline{Y}$ is called the normal α-stable distribution on $\mathbb{R}^{d-1}$ and its Laplace transform is given by*

$$L_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta}) = e^{1-\left(1-\beta\frac{\|\overline{\theta}\|^2}{2}\right)^{\alpha}}, \text{ for all } \overline{\theta} \in B_{d-1}(0,\sqrt{2/\beta}),$$

where $B_{d-1}(0,\sqrt{2/\beta})$ is the ball in $\mathbb{R}^{d-1}$ centered at 0, with a radius $\sqrt{2/\beta}$.

Proof

For all $\theta \in \left\{(\theta_1,\overline{\theta}) \in \mathbb{R}\times\mathbb{R}^{d-1}\ ;\ \theta_1 + \frac{\|\overline{\theta}\|^2}{2} < 1/\beta\right\}$,

$$\begin{aligned}
L_{\mu_{\alpha,\beta}}(\theta) &= \mathbb{E}(e^{\langle\theta,Y\rangle}) = \mathbb{E}(e^{\theta_1 Y_1 + \langle\overline{\theta},\overline{Y}\rangle}) \\
&= \int_0^{+\infty}\int_{\mathbb{R}^{d-1}} e^{\theta_1 y_1} e^{\langle\overline{\theta},\overline{y}\rangle} N(0,y_1 I_{d-1})(d\overline{y}) \otimes \pi_{\alpha,\beta}(dy_1) \\
&= \int_0^{+\infty} e^{\theta_1 y_1}\left(\int_{\mathbb{R}^{d-1}} e^{\langle\overline{\theta},\overline{y}\rangle} N(0,y_1 I_{d-1})(d\overline{y})\right)\pi_{\alpha,\beta}(dy_1) \\
&= \int_0^{+\infty} e^{\theta_1 y_1} e^{\frac{y_1\|\overline{\theta}\|^2}{2}}\pi_{\alpha,\beta}(dy_1) \\
&= \int_0^{+\infty} e^{y_1\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)}\pi_{\alpha,\beta}(dy_1) \\
&= L_{\pi_{\alpha,\beta}}\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right) \\
&= e^{1-\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{\alpha}}. \qquad (2.1)
\end{aligned}$$

Taking $\theta_1 = 0$ in (2.1), we deduce that, for all $\theta = (0,\overline{\theta})$ such that $\overline{\theta} \in B_{d-1}(0,\sqrt{2/\beta})$,

$$L_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta}) = \mathbb{E}\left(e^{\langle\overline{\theta},\overline{Y}\rangle}\right) = L_{\mu_{\alpha,\beta}}(0,\overline{\theta}) = e^{1-\left(1-\beta\frac{\|\overline{\theta}\|^2}{2}\right)^{\alpha}}. \qquad (2.2)$$

□

It is important to notice that when $\alpha = 1/2$, $\overline{\mu}_{1/2,\beta}$ corresponds to the normal inverse Gaussian distribution (see Barndorff-Nielsen (1997)), and in this case (2.2) becomes

$$L_{\overline{\mu}_{1/2,\beta}}(\overline{\theta}) = e^{1-\sqrt{1-\beta\frac{\|\overline{\theta}\|^2}{2}}}.$$

2.2 Variance function

In this section, we determine explicitly the expression of the variance function of the NEF $F(\overline{\mu}_{\alpha,\beta})$ generated by the normal α-stable distribution. For this purpose, let's give the expression of the Hessian of the cumulant function of $\mu_{\alpha,\beta}$.

Proposition 2.2.1

1. The covariance operator of $\mu_{\alpha,\beta}$ is given, for all $\theta \in \{(\theta_1,\overline{\theta}) \in \mathbb{R}^d \ ; \ \theta_1 + \frac{\|\overline{\theta}\|^2}{2} < 1/\beta\}$, by

$$k''_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta}) = \frac{\alpha(1-\alpha)\beta^2(1,\overline{\theta})\otimes(1,\overline{\theta})}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{2-\alpha}} + \frac{\alpha\beta\left(I_d - e_1\otimes e_1\right)}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{1-\alpha}}.$$

2. The covariance operator of $\overline{\mu}_{\alpha,\beta}$ is given, for all $\overline{\theta} \in \Theta(\overline{\mu}_{\alpha,\beta}) = B_{d-1}(0,\sqrt{2/\beta})$, by

$$k''_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta}) = \frac{\alpha\beta}{\left(1-\beta\frac{\|\overline{\theta}\|^2}{2}\right)^{1-\alpha}} I_{d-1} + \frac{\alpha(1-\alpha)\beta^2}{\left(1-\beta\frac{\|\overline{\theta}\|^2}{2}\right)^{2-\alpha}}\overline{\theta}\otimes\overline{\theta}.$$

Proof

1. According to (2.1), we have, for all $\theta \in \{(\theta_1,\overline{\theta}) \in \mathbb{R}^d;\ \theta_1 + \frac{\|\overline{\theta}\|^2}{2} < 1/\beta\}$,

$$k_{\mu_{\alpha,\beta}}(\theta) = k_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta}) = 1 - \left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{\alpha}. \tag{2.3}$$

The derivatives of $k_{\mu_{\alpha,\beta}}(\theta)$ with respect to θ_1 and $\overline{\theta}$ are given by

$$\frac{\partial k_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta})}{\partial\theta_1} = \alpha\beta\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{\alpha-1}, \tag{2.4}$$

and

$$\frac{\partial k_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta})}{\partial\overline{\theta}} = \alpha\beta\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{\alpha-1}\overline{\theta}. \tag{2.5}$$

Differentiating these with respect to θ_1 and $\overline{\theta}$, we get

$$\frac{\partial^2 k_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta})}{\partial\theta_1^2} = \frac{\alpha(1-\alpha)\beta^2}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{2-\alpha}}, \tag{2.6}$$

$$\frac{\partial^2 k_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta})}{\partial\theta_1\partial\overline{\theta}} = \frac{\alpha(1-\alpha)\beta^2}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{2-\alpha}}\overline{\theta}, \tag{2.7}$$

and

$$\frac{\partial^2 k_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta})}{\partial\overline{\theta}^2}=\frac{\alpha\beta}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{1-\alpha}}I_{d-1}+\frac{\alpha(1-\alpha)\beta^2}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{2-\alpha}}\overline{\theta}\otimes\overline{\theta}. \tag{2.8}$$

Therefore, the covariance operator $k''_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta})$ is equal to

$$\frac{\alpha(1-\alpha)\beta^2}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{2-\alpha}}\begin{pmatrix}1 & \overline{\theta}\\ \overline{\theta}' & \overline{\theta}\otimes\overline{\theta}\end{pmatrix}+\frac{\alpha\beta}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{1-\alpha}}\begin{pmatrix}0 & 0\\ 0 & I_{d-1}\end{pmatrix}, \tag{2.9}$$

where $\overline{\theta}'$ denotes the vector transpose of $\overline{\theta}$. This implies that

$$k''_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta})=\frac{\alpha(1-\alpha)\beta^2(1,\overline{\theta})\otimes(1,\overline{\theta})}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{2-\alpha}}+\frac{\alpha\beta\left(I_d-e_1\otimes e_1\right)}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{1-\alpha}}.$$

2. Using (2.3) and the fact that

$$k_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta})=k_{\mu_{\alpha,\beta}}(0,\overline{\theta}), \tag{2.10}$$

we deduce that, for all $\overline{\theta}\in B_{d-1}(0,\sqrt{2/\beta})$,

$$k_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta})=1-\left(1-\beta\frac{\|\overline{\theta}\|^2}{2}\right)^{\alpha}.$$

Differentiating this twice with respect to $\overline{\theta}$, we get the desired result. □

Remark 2.2.2 *The random vector* $\dfrac{\overline{Y}}{\sqrt{\alpha\beta}}$ *is standard normal α-stable distributed. In fact, using (2.5) and (2.10), we get*

$$\mathbb{E}(\frac{\overline{Y}}{\sqrt{\alpha\beta}})=\frac{k'_{\overline{\mu}_{\alpha,\beta}}(0)}{\sqrt{\alpha\beta}}=0.$$

Furthermore, using (2.9), we deduce that the variance of the random variable $\frac{\overline{Y}}{\sqrt{\alpha\beta}}$ *is*

$$Var\left(\frac{\overline{Y}}{\sqrt{\alpha\beta}}\right)=\frac{k''_{\mu_{\alpha,\beta}}(0)}{\alpha\beta}=I_{d-1}.$$

Next, we give the variance function of the NEF $F(\mu_{\alpha,\beta})$ generated by $\mu_{\alpha,\beta}$.

Theorem 2.2.3 *For all $m = (m_1, \overline{m}) \in (0, \infty) \times \mathbb{R}^{d-1}$, the variance function of $F(\mu_{\alpha,\beta})$ is given by*

$$V_{F(\mu_{\alpha,\beta})}(m_1, \overline{m}) = \alpha^{\frac{1}{\alpha-1}}(1-\alpha)\beta^{\frac{\alpha}{\alpha-1}} m_1^{\frac{\alpha}{1-\alpha}} m \otimes m + m_1 \left(I_d - e_1 \otimes e_1\right). \tag{2.11}$$

Proof

According to (2.4) and (2.5), we have

$$k'_{\mu_{\alpha,\beta}}(\theta_1, \overline{\theta}) = \left(\frac{\partial k_{\mu_{\alpha,\beta}}(\theta_1, \overline{\theta})}{\partial \theta_1}, \frac{\partial k_{\mu_{\alpha,\beta}}(\theta_1, \overline{\theta})}{\partial \overline{\theta}} \right) = m = (m_1, \overline{m}) \in (0, \infty) \times \mathbb{R}^{d-1}.$$

Then, using (2.4), we deduce that

$$1 - \beta\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right) = \left(\frac{m_1}{\alpha\beta}\right)^{\frac{1}{\alpha-1}}.$$

According to (2.5), we have

$$\overline{\theta} = \frac{\overline{m}}{m_1}.$$

Inserting this in (2.9), we get

$$V_{F(\mu_{\alpha,\beta})}(m_1, \overline{m}) = \alpha^{\frac{1}{\alpha-1}}(1-\alpha)\beta^{\frac{\alpha}{\alpha-1}} m_1^{\frac{\alpha}{1-\alpha}} \begin{pmatrix} m_1^2 & m_1\overline{m} \\ m_1\overline{m}' & \overline{m} \otimes \overline{m} \end{pmatrix} + m_1 \begin{pmatrix} 0 & 0 \\ 0 & I_{d-1} \end{pmatrix}. \tag{2.12}$$

Hence, the variance function of $F(\mu_{\alpha,\beta})$ is given, for all $m \in (0, \infty) \times \mathbb{R}^{d-1}$, by $V_{F(\mu_{\alpha,\beta})}(m_1, \overline{m}) = \alpha^{\frac{1}{\alpha-1}}(1-\alpha)\beta^{\frac{\alpha}{\alpha-1}} m_1^{\frac{\alpha}{1-\alpha}} m \otimes m + m_1 \left(I_d - e_1 \otimes e_1\right)$. □

Remark 2.2.4 *The particular case where $\alpha = 1/2$ corresponds to the inverse Gaussian family introduced by Hassairi (1999). In fact, let $Y = (Y_1, \overline{Y})$ be a random vector such that Y_1 is inverse Gaussian distributed and $\overline{Y}$ is normal inverse Gaussian distributed. Taking $\alpha = 1/2$ in (2.11), we obtain, for all $m = (m_1, \overline{m}) \in (0, \infty) \times \mathbb{R}^{d-1}$,*

$$V_{F(\mu_{1/2,\beta})}(m_1, \overline{m}) = \frac{2}{\beta} m_1 m \otimes m + m_1(I_d - e_1 \otimes e_1).$$

This corresponds to a multivariate cubic variance function (see Hassairi and Zarai (2004)). For $d = 2$, we obtain, for all $m = (m_1, \overline{m}) \in (0, \infty) \times \mathbb{R}$,

$$V_{F(\mu_{1/2,\beta})} = \begin{pmatrix} \frac{2}{\beta} m_1^3 & \frac{2}{\beta} m_1^2 \overline{m} \\ \frac{2}{\beta} m_1^2 \overline{m} & \frac{2}{\beta} m_1 \overline{m}^2 + m_1 \end{pmatrix}.$$

Now, we characterize the natural exponential family $F(\overline{\mu}_{\alpha,\beta})$ generated by the normal α-stable distribution by means its variance function. More precisely, we have

Theorem 2.2.5 *The variance function of $F(\overline{\mu}_{\alpha,\beta})$ is given by*

$$V_{F(\overline{\mu}_{\alpha,\beta})}(\overline{m}) = \xi(\overline{m})I_{d-1} + \alpha^{\frac{1}{\alpha-1}}(1-\alpha)\beta^{\frac{\alpha}{\alpha-1}}\xi(\overline{m})^{\frac{\alpha}{1-\alpha}}\overline{m}\otimes\overline{m}, \text{ for all } \overline{m}\in\mathbb{R}^{d-1},$$

where $\xi(\overline{m})$ is the unique solution of the equation

$$(\alpha\beta)^{\frac{1}{1-\alpha}}s^{\frac{2\alpha-1}{\alpha-1}} - s^2 + \frac{\beta\|\overline{m}\|^2}{2} = 0, \text{ for all } s>0.$$

Before starting the proof of this theorem, we need to study and prove the two following technical lemmas.

Lemma 2.2.6 *Let α be an element of $(0,1)$ and let β be a strictly positive real. Then, the function ϕ defined on $(0,+\infty)$ by*

$$\phi(x) = (\alpha\beta)^{\frac{1}{1-\alpha}}x^{\frac{2\alpha-1}{\alpha-1}} - x^2 + \frac{\beta\|\overline{m}\|^2}{2} \tag{2.13}$$

admits a unique positive solution $\xi(\overline{m})$.

Proof of Lemma 2.2.6

Differentiating ϕ with respect to x, we obtain, for all $x>0$,

$$\phi'(x) = (\alpha\beta)^{\frac{1}{1-\alpha}}\frac{2\alpha-1}{\alpha-1}x^{\frac{\alpha}{\alpha-1}} - 2x. \tag{2.14}$$

We distinguish three cases.

Case 1: $\alpha = 1/2$

In this case, ϕ is strictly decreasing from $(0,+\infty)$ to $\phi((0,+\infty)) = (-\infty, \frac{\beta^2}{4} + \frac{\beta\|\overline{m}\|^2}{2})$. Since $\frac{\beta^2}{4} + \frac{\beta\|\overline{m}\|^2}{2} > 0$, we deduce that ϕ admits a unique positive zero.

Case 2: $0 < \alpha < 1/2$

In this case,

$$\phi'(x) = 0 \Leftrightarrow x = x_0 = \alpha\beta\left(\frac{2\alpha-1}{2\alpha-2}\right)^{1-\alpha} > 0.$$

By using the fact that ϕ' admits a unique zero x_0 in $(0,+\infty)$, $\lim\limits_{x\mapsto 0}\phi'(x)=+\infty$ and $\lim\limits_{x\mapsto+\infty}\phi'(x)=-\infty$, we deduce that

$$\phi'(x)>0,\ \text{for all } x\in(0,x_0)$$

and

$$\phi'(x)<0,\ \text{for all } x\in(x_0,+\infty).$$

This implies that the function ϕ is strictly increasing on $(0,x_0)$ and strictly decreasing on $(x_0,+\infty)$. Furthermore, since $\phi(0)=\frac{\beta\|\overline{m}\|^2}{2}>0$, then $\phi(x_0)>0$, and since $\lim\limits_{x\mapsto+\infty}\phi(x)=-\infty$, we conclude that ϕ has a unique positive zero $\xi(\overline{m})>x_0$.

Case 3: $1/2<\alpha<1$

Using (2.14), we deduce that ϕ is strictly decreasing from $(0,+\infty)$ to $\phi((0,+\infty))=\mathbb{R}$. It follows that the equation $\phi(x)=0$ admits a unique solution. □

In this lemma, we give explicitly the first and the second derivative of the cumulant function $k_{\mu_{\alpha,\beta}}$ of $\mu_{\alpha,\beta}$.

Lemma 2.2.7 *For all $\overline{\theta}\in B_{d-1}(0,\sqrt{\beta/2})$, we have*

$$k'_{\mu_{\alpha,\beta}}(0,\overline{\theta})=\left(\frac{\partial k_{\mu_{\alpha,\beta}}(0,\overline{\theta})}{\partial\theta_1},k'_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta})\right),$$

and

$$k''_{\mu_{\alpha,\beta}}(0,\overline{\theta})=\begin{pmatrix}\frac{\partial^2 k_{\mu_{\alpha,\beta}}(0,\overline{\theta})}{\partial\theta_1^2} & \frac{\partial^2 k_{\mu_{\alpha,\beta}}(0,\overline{\theta})}{\partial\overline{\theta}\partial\theta_1}\\ \left(\frac{\partial^2 k_{\mu_{\alpha,\beta}}(0,\overline{\theta})}{\partial\theta_1\partial\overline{\theta}}\right)' & k''_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta})\end{pmatrix}.$$

Proof of Lemma 2.2.7

According to (1.1), we have for all $(\theta_1,\overline{\theta})\in\Theta(\mu_{\alpha,\beta})=\{(\theta_1,\overline{\theta})\in\mathbb{R}^d\ ;\ \theta_1+\frac{\|\overline{\theta}\|^2}{2}<1/\beta\}$

$$k'_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta})=\int_0^{+\infty}\int_{\mathbb{R}^{d-1}}(y_1,\overline{y})P((\theta_1,\overline{\theta}),\mu_{\alpha,\beta})(dy_1,d\overline{y})\tag{2.15}$$

It follows that, for all $(\theta_1,\overline{\theta})\in\Theta(\mu_{\alpha,\beta})$,

$$k'_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta})=\left(\int_0^{+\infty}\int_{\mathbb{R}^{d-1}}y_1P((\theta_1,\overline{\theta}),\mu_{\alpha,\beta})(dy_1,d\overline{y}),\int_0^{+\infty}\int_{\mathbb{R}^{d-1}}\overline{y}P((\theta_1,\overline{\theta}),\mu_{\alpha,\beta})(dy_1,d\overline{y})\right).$$

Setting $\theta_1 = 0$ and using the fact that

$$P((0,\overline{\theta}),\mu_{\alpha,\beta})(dy_1,d\overline{y}) = e^{\langle \overline{\theta},\overline{y}\rangle - k_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta})}\mu_{\alpha,\beta}(dy_1,d\overline{y}),$$

we deduce that, for all $\overline{\theta} \in B_{d-1}(0,\sqrt{\beta/2})$,

$$\begin{aligned}
\frac{\partial k_{\mu_{\alpha,\beta}}(0,\overline{\theta})}{\partial \overline{\theta}} &= \int_0^{+\infty}\int_{\mathbb{R}^{d-1}} \overline{y}P((0,\overline{\theta}),\mu_{\alpha,\beta})(dy_1,d\overline{y}) \\
&= \int_0^{+\infty}\int_{\mathbb{R}^{d-1}} \overline{y}e^{\langle \overline{\theta},\overline{y}\rangle - k_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta})}\mu_{\alpha,\beta}(dy_1,d\overline{y}) \\
&= \int_{\mathbb{R}^{d-1}} \overline{y}e^{\langle \overline{\theta},\overline{y}\rangle - k_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta})}\overline{\mu}_{\alpha,\beta}(d\overline{y}) \\
&= k'_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta}). \qquad (2.16)
\end{aligned}$$

Inserting this in (2.15), we conclude that, for all $\overline{\theta} \in B_{d-1}(0,\sqrt{\beta/2})$,

$$\begin{aligned}
k'_{\mu_{\alpha,\beta}}(0,\overline{\theta}) &= \left(\int_0^{+\infty}\int_{\mathbb{R}^{d-1}} y_1 P((0,\overline{\theta}),\mu_{\alpha,\beta})(dy_1,d\overline{y}), \int_0^{+\infty}\int_{\mathbb{R}^{d-1}} \overline{y}P((0,\overline{\theta}),\mu_{\alpha,\beta})(dy_1,d\overline{y})\right) \\
&= \left(\int_0^{+\infty}\int_{\mathbb{R}^{d-1}} y_1 P((0,\overline{\theta}),\mu_{\alpha,\beta})(dy_1,d\overline{y}), k'_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta})\right) \\
&= \left(\frac{\partial k_{\mu_{\alpha,\beta}}(0,\overline{\theta})}{\partial \theta_1}, k'_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta})\right).
\end{aligned}$$

Furthermore, differentiating (2.16) with respect to $\overline{\theta}$, we can write

$$\begin{aligned}
k''_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta}) &= \int_0^{+\infty}\int_{\mathbb{R}^{d-1}} \overline{y}\otimes\overline{y}P((0,\overline{\theta}),\mu_{\alpha,\beta})(dy_1,d\overline{y}) - k'_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta})\otimes k'_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta}) \\
&= \frac{\partial^2 k_{\mu_{\alpha,\beta}}(0,\overline{\theta})}{\partial \overline{\theta}^2}.
\end{aligned}$$

This represents the desired result. □

Now, we are in position to prove Theorem 3.2.1.

Proof of Theorem 3.2.1

Using (2.4) and (2.5), we deduce that, for a fixed $\theta_1 = 0$ and for $\overline{\theta} \in B_{d-1}(0,\sqrt{2/\beta})$, there exist $m_1(0,\overline{\theta}) \in (0,+\infty)$ and $\overline{m} \in \mathbb{R}^{d-1}$, such that

$$\frac{\partial k_{\mu_{\alpha,\beta}}(0,\overline{\theta})}{\partial \theta_1} = \alpha\beta\left(1-\beta\frac{\|\overline{\theta}\|^2}{2}\right)^{\alpha-1} = m_1(0,\overline{\theta}) = m_1 > 0,$$

and

$$\frac{\partial k_{\mu_{\alpha,\beta}}(0,\overline{\theta})}{\partial\overline{\theta}} = \alpha\beta\left(1-\beta\frac{\|\overline{\theta}\|^2}{2}\right)^{\alpha-1}\overline{\theta} = \overline{m}(0,\overline{\theta}) = \overline{m} \in \mathbb{R}^{d-1}.$$

Therefore,

$$m_1 = \alpha\beta\left(1-\beta\frac{\|\overline{m}\|^2}{2m_1^2}\right)^{\alpha-1}.$$

It follows that

$$(\alpha\beta)^{\frac{1}{1-\alpha}} m_1^{\frac{2\alpha-1}{\alpha-1}} - m_1^2 + \frac{\beta\|\overline{m}\|^2}{2} = 0. \tag{2.17}$$

Hence, m_1 is a positive zero of the function ϕ defined in (2.13). This, together with Lemma 2.2.6, imply that

$$m_1 = \xi(\overline{m}) \tag{2.18}$$

Furthermore, by combining Lemma 2.2.7 with the equality (2.12), we deduce that

$$k''_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta}) = V_{F(\overline{\mu}_{\alpha,\beta})}(\overline{m}) = m_1 I_{d-1} + \alpha^{\frac{1}{\alpha-1}}(1-\alpha)\beta^{\frac{\alpha}{\alpha-1}} m_1^{\frac{\alpha}{1-\alpha}}\overline{m}\otimes\overline{m}. \tag{2.19}$$

Inserting (2.18) in (2.19), we get for all $\overline{m} \in \mathbb{R}^{d-1}$,

$$V_{F(\overline{\mu}_{\alpha,\beta})}(\overline{m}) = \xi(\overline{m}) I_{d-1} + \alpha^{\frac{1}{\alpha-1}}(1-\alpha)\beta^{\frac{\alpha}{\alpha-1}} \xi(\overline{m})^{\frac{\alpha}{1-\alpha}}\overline{m}\otimes\overline{m}. \tag{2.20}$$

□

Remark 2.2.8 *Note that, if $\alpha = 1/2$, then the solution of (2.17) is*

$$\xi(\overline{m}) = \sqrt{\frac{\beta\|\overline{m}\|^2}{2} + \frac{\beta^2}{4}}.$$

In this case, (2.20) becomes

$$V_{F(\overline{\mu}_{1/2,\beta})}(\overline{m}) = \sqrt{\frac{\beta\|\overline{m}\|^2}{2} + \frac{\beta^2}{4}}\left(I_{d-1} + \frac{2}{\beta}\overline{m}\otimes\overline{m}\right), \text{ for all } \overline{m} \in \mathbb{R}^{d-1}.$$

This implies that, $F(\overline{\mu}_{1/2,\beta})$ is the normal inverse Gaussian family on $\mathbb{R}^{d-1}$.
For $d = 2$, we get

$$V_{F(\overline{\mu}_{1/2,\beta})}(\overline{m}) = \sqrt{\frac{\beta\overline{m}^2}{2} + \frac{\beta^2}{4}}\left(1 + \frac{2}{\beta}\overline{m}^2\right), \text{ for all } \overline{m} \in \mathbb{R}.$$

2.3 Lévy measures

In this section, we determine the Lévy measures of the distribution $\mu_{\alpha,\beta}$ and the normal α-stable distribution $\overline{\mu}_{\alpha,\beta}$ which are infinitely divisible (see Letac and Seshadri (2005)). For this purpose, we use the Lévy-Khintchine representation. It is important to notice that Seshadri (1994) has obtained, in one dimension, the Lévy measure associated to the class of stable distributions.

Lemma 2.3.1 *There exists a Lévy measure $\nu_{\alpha,\beta}$ such that*

$$k''_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta}) = \int_{\mathbb{R}^d\setminus\{0\}} (y_1,\overline{y})\otimes(y_1,\overline{y}) e^{\theta_1 y_1+\langle\overline{\theta},\overline{y}\rangle}\nu_{\alpha,\beta}(dy_1,d\overline{y}).$$

Proof

According to Proposition 2.2.1, we have for all $\theta\in\{(\theta_1,\overline{\theta})\in\mathbb{R}^d\ ;\ \theta_1+\frac{\|\overline{\theta}\|^2}{2}<1/\beta\}$,

$$k''_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta}) = \frac{\alpha(1-\alpha)\beta^2}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{2-\alpha}}(1,\overline{\theta})\otimes(1,\overline{\theta}) + \frac{\alpha\beta}{\left(1-\beta\left(\theta_1+\frac{\|\overline{\theta}\|^2}{2}\right)\right)^{1-\alpha}}(I_d-e_1\otimes e_1).$$

It follows that

$$\lim_{\theta_1\mapsto-\infty} k''_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta}) = 0. \tag{2.21}$$

Furthermore, since $\mu_{\alpha,\beta}$ is infinitely divisible, and according to Lévy-Khintchine representation, there exist a symmetric positive matrix Σ and a positive measure $\nu_{\alpha,\beta}$ such that

$$k''_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta}) = \Sigma + \int_{\mathbb{R}^d\setminus\{0\}} (y_1,\overline{y})\otimes(y_1,\overline{y}) e^{\theta_1 y_1+\langle\overline{\theta},\overline{y}\rangle}\nu_{\alpha,\beta}(dy_1,d\overline{y}). \tag{2.22}$$

This, combined with (2.21), imply that

$$0 = \lim_{\theta_1\mapsto-\infty} k''_{\mu_{\alpha,\beta}}(\theta_1,\overline{\theta}) = \Sigma + \lim_{\theta_1\mapsto-\infty}\int_{\mathbb{R}^d\setminus\{0\}} (y_1,\overline{y})\otimes(y_1,\overline{y}) e^{\theta_1 y_1+\langle\overline{\theta},\overline{y}\rangle}\nu_{\alpha,\beta}(dy_1,d\overline{y}).$$

Since Σ and $\int_{\mathbb{R}^d\setminus\{0\}}(y_1,\overline{y})\otimes(y_1,\overline{y})e^{\theta_1 y_1+\langle\overline{\theta},\overline{y}\rangle}\nu_{\alpha,\beta}(dy_1,d\overline{y})$ are positive matrices, then we deduce that

$$\Sigma = 0.$$

Inserting this in (2.22), we get the result. □

Next, we give the Lévy measure of $\mu_{\alpha,\beta}$.

Theorem 2.3.2 *The Lévy measure of $\mu_{\alpha,\beta}$ is*

$$\nu_{\alpha,\beta}(dy_1, d\overline{y}) = \frac{\alpha\beta}{y_1} N(0, y_1 I_{d-1})(d\overline{y}) \otimes \gamma(1-\alpha, \beta)(dy_1), \tag{2.23}$$

where $\gamma(1-\alpha,\beta)$ is the real Gamma distribution with a shape parameter $1-\alpha > 0$ and a scale parameter $\beta > 0$.

Proof

Using (2.6), we have for all $\theta \in \left\{(\theta_1, \overline{\theta}) \in \mathbb{R} \times \mathbb{R}^{d-1} \; ; \; \theta_1 + \frac{\|\overline{\theta}\|^2}{2} < 1/\beta\right\}$,

$$\begin{aligned}
\frac{\partial^2 k_{\mu_{\alpha,\beta}}(\theta_1, \overline{\theta})}{\partial \theta_1^2} &= \frac{\alpha(1-\alpha)\beta^2}{\left(1 - \beta\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right)\right)^{2-\alpha}} \\
&= \alpha\beta \int_0^{+\infty} y_1 e^{y_1\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right)} \gamma(1-\alpha,\beta)(dy_1) \\
&= \alpha\beta \int_0^{+\infty} y_1 e^{\theta_1 y_1} \left(\int_{\mathbb{R}^{d-1}} e^{\langle \overline{\theta}, \overline{y}\rangle} N(0, y_1 I_{d-1})(d\overline{y}) \right) \gamma(1-\alpha,\beta)(dy_1) \\
&= \int_0^{+\infty} \int_{\mathbb{R}^{d-1}} y_1^2 e^{\theta_1 y_1 + \langle \overline{\theta}, \overline{y}\rangle} \frac{\alpha\beta}{y_1} N(0, y_1 I_{d-1})(d\overline{y}) \otimes \gamma(1-\alpha,\beta)(dy_1).
\end{aligned}$$

It follows that

$$\frac{\partial^2 k_{\mu_{\alpha,\beta}}(\theta_1, \overline{\theta})}{\partial \theta_1^2} = \int_{\mathbb{R}^d \setminus \{0\}} y_1^2 e^{\theta_1 y_1 + \langle \overline{\theta}, \overline{y}\rangle} \nu_{\alpha,\beta}(dy_1, d\overline{y}), \tag{2.24}$$

where the measure $\nu_{\alpha,\beta}$ is given by (2.23). Moreover, according to (2.7), we have

$$\begin{aligned}
\frac{\partial^2 k_{\mu_{\alpha,\beta}}(\theta_1, \overline{\theta})}{\partial \theta_1 \partial \overline{\theta}} &= \frac{\alpha(1-\alpha)\beta^2}{\left(1 - \beta\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right)\right)^{2-\alpha}} \overline{\theta} \\
&= \alpha\beta\overline{\theta} \int_0^{+\infty} y_1 e^{y_1\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right)} \gamma(1-\alpha,\beta)(dy_1) \\
&= \alpha\beta \int_0^{+\infty} e^{\theta_1 y_1} \left(\int_{\mathbb{R}^{d-1}} \overline{y} e^{\langle \overline{\theta}, \overline{y}\rangle} N(0, y_1 I_{d-1})(d\overline{y}) \right) \gamma(1-\alpha,\beta)(dy_1) \\
&\quad \int_0^{+\infty} \int_{\mathbb{R}^{d-1}} y_1 \overline{y} e^{\theta_1 y_1 + \langle \overline{\theta}, \overline{y}\rangle} \frac{\alpha\beta}{y_1} N(0, y_1 I_{d-1})(d\overline{y}) \otimes \gamma(1-\alpha,\beta)(dy_1).
\end{aligned}$$

This implies that

$$\frac{\partial^2 k_{\mu_{\alpha,\beta}}(\theta_1, \overline{\theta})}{\partial \theta_1 \partial \overline{\theta}} = \int_{\mathbb{R}^d \setminus \{0\}} y_1 \overline{y} e^{\theta_1 y_1 + \langle \overline{\theta}, \overline{y}\rangle} \nu_{\alpha,\beta}(dy_1, d\overline{y}). \tag{2.25}$$

Finally, using (2.8), we get for all $\theta \in \left\{(\theta_1, \overline{\theta}) \in \mathbb{R} \times \mathbb{R}^{d-1} \; ; \; \theta_1 + \frac{\|\overline{\theta}\|^2}{2} < 1/\beta\right\}$,

$$\begin{aligned}
\frac{\partial^2 k_{\mu_{\alpha,\beta}}(\theta_1, \overline{\theta})}{\partial \overline{\theta}^2} &= \frac{\alpha\beta}{\left(1 - \beta\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right)\right)^{1-\alpha}} I_{d-1} + \frac{\alpha(1-\alpha)\beta^2}{\left(1 - \beta\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right)\right)^{2-\alpha}} \overline{\theta} \otimes \overline{\theta} \\
&= \alpha\beta \int_0^{+\infty} e^{y_1\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right)} \gamma(1-\alpha, \beta)(dy_1) I_{d-1} \\
&+ \alpha\beta \int_0^{+\infty} y_1 e^{y_1\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right)} \gamma(1-\alpha, \beta)(dy_1) \overline{\theta} \otimes \overline{\theta} \\
&= \alpha\beta \int_0^{+\infty} e^{y_1\left(\theta_1 + \frac{\|\overline{\theta}\|^2}{2}\right)} (I_{d-1} + y_1 \overline{\theta} \otimes \overline{\theta}) \gamma(1-\alpha, \beta)(dy_1) \\
&= \alpha\beta \int_0^{+\infty} \frac{1}{y_1} e^{\theta_1 y_1} \left(\int_{\mathbb{R}^{d-1}} \overline{y} \otimes \overline{y} e^{\langle \overline{\theta}, \overline{y} \rangle} N(0, y_1 I_{d-1})(d\overline{y}) \right) \gamma(1-\alpha, \beta)(dy_1) \\
&= \int_0^{+\infty} \int_{\mathbb{R}^{d-1}} \overline{y} \otimes \overline{y} e^{\theta_1 y_1 + \langle \overline{\theta}, \overline{y} \rangle} \frac{\alpha\beta}{y_1} N(0, y_1 I_{d-1})(d\overline{y}) \otimes \gamma(1-\alpha, \beta)(dy_1).
\end{aligned}$$

Hence, for all $\theta \in \left\{(\theta_1, \overline{\theta}) \in \mathbb{R} \times \mathbb{R}^{d-1} \; ; \; \theta_1 + \frac{\|\overline{\theta}\|^2}{2} < 1/\beta\right\}$,

$$\frac{\partial^2 k_{\mu_{\alpha,\beta}}(\theta_1, \overline{\theta})}{\partial \overline{\theta}^2} = \int_{\mathbb{R}^d \setminus \{0\}} \overline{y} \otimes \overline{y} e^{\theta_1 y_1 + \langle \overline{\theta}, \overline{y} \rangle} \nu_{\alpha,\beta}(dy_1, d\overline{y}). \tag{2.26}$$

Consequently, (2.24), (2.25) and (2.26) give

$$k''_{\mu_{\alpha,\beta}}(\theta_1, \overline{\theta}) = \int_{\mathbb{R}^d \setminus \{0\}} (y_1, \overline{y}) \otimes (y_1, \overline{y}) e^{\theta_1 y_1 + \langle \overline{y}, \overline{\theta} \rangle} \frac{\alpha\beta}{y_1} N(0, y_1 I_{d-1})(d\overline{y}) \otimes \gamma(1-\alpha, \beta)(dy_1).$$

This represents the desired result. □

We illustrate the result of the previous theorem by an example.

Example 2.3.3 *For $\alpha = 1/2$, the Lévy measure of the distribution $\mu_{1/2,\beta}$ is given by*

$$\nu_{1/2,\beta}(dy_1, d\overline{y}) = \frac{\beta}{2y_1} N(0, y_1 I_{d-1})(d\overline{y}) \otimes \gamma(1/2, \beta)(dy_1).$$

For $d = 2$, we get

$$\nu_{1/2,\beta}(dy_1, d\overline{y}) = \frac{\beta}{2y_1} N(0, y_1)(d\overline{y}) \otimes \gamma(1/2, \beta)(dy_1).$$

Next, we give the Lévy measure of $\overline{\mu}_{\alpha,\beta}$.

Theorem 2.3.4 *The Lévy measure of $\overline{\mu}_{\alpha,\beta}$ is*

$$\overline{\nu}_{\alpha,\beta}(d\overline{y}) = \frac{\alpha\beta}{\Gamma(\beta)2\beta^{\alpha}(2\pi)^{\frac{d-1}{2}}}\left(\frac{2}{\beta\|\overline{y}\|^2}\right)^{\frac{2-d}{4}-\frac{\alpha}{2}} K_{\frac{2-d}{2}-\alpha}\left(\sqrt{2\|\overline{y}\|^2/\beta}\right)d\overline{y}, \tag{2.27}$$

where $K_{\frac{2-d}{2}-\alpha}(.)$ denotes the modified Bessel function of the third kind with order $\frac{2-d}{2}-\alpha$ defined in (1.8).

Proof

According to Lemma 2.2.7 and Lemma 2.3.1, if we take $\theta_1 = 0$, we obtain

$$k''_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta}) = \int_{\mathbb{R}^d\setminus\{0\}} \overline{y}\otimes\overline{y}e^{\langle\overline{\theta},\overline{y}\rangle}\nu_{\alpha,\beta}(dy_1,\overline{y}).$$

Combined with (2.23), this gives

$$\begin{aligned}
k''_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta}) &= \int_{\mathbb{R}^d\setminus\{0\}} \overline{y}\otimes\overline{y}e^{\langle\overline{\theta},\overline{y}\rangle}\frac{\alpha\beta}{y_1}N(0,y_1I_{d-1})(d\overline{y})\otimes\gamma(1-\alpha,\beta)(dy_1)\\
&= \int_{\mathbb{R}^{d-1}} \overline{y}\otimes\overline{y}e^{\langle\overline{\theta},\overline{y}\rangle}\left(\frac{\alpha\beta}{\Gamma(\beta)\beta^{\alpha}(2\pi)^{\frac{d-1}{2}}}\int_0^{+\infty}\frac{e^{-\frac{\|\overline{y}\|^2}{2y_1}}e^{-\frac{y_1}{\beta}}}{y_1^{\frac{d+1}{2}+\alpha}}dy_1\right)d\overline{y}\\
&= \int_{\mathbb{R}^{d-1}} \overline{y}\otimes\overline{y}e^{\langle\overline{\theta},\overline{y}\rangle}\frac{\alpha\beta}{\Gamma(\beta)2\beta^{\alpha}(2\pi)^{\frac{d-1}{2}}}\left(\frac{2}{\beta\|\overline{y}\|^2}\right)^{-\frac{d-2}{4}-\frac{\alpha}{2}} K_{\frac{2-d}{2}-\alpha}\left(\sqrt{2\|\overline{y}\|^2/\beta}\right)d\overline{y}.
\end{aligned}$$

Therefore,

$$k''_{\overline{\mu}_{\alpha,\beta}}(\overline{\theta}) = \int_{\mathbb{R}^{d-1}} \overline{y}\otimes\overline{y}e^{\langle\overline{\theta},\overline{y}\rangle}\overline{\nu}_{\alpha,\beta}(d\overline{y}),$$

where the measure $\overline{\nu}_{\alpha,\beta}$ is defined in (2.27). □

Finally, we illustrate the result of the previous theorem by an example.

Example 2.3.5 *For $\alpha = 1/2$, the Lévy measure of the normal inverse Gaussian distribution $\overline{\mu}_{1/2,\beta}$ is given by*

$$\overline{\nu}_{1/2,\beta}(d\overline{y}) = \frac{\sqrt{\beta}}{4\Gamma(\beta)(2\pi)^{\frac{d-1}{2}}}\left(\frac{2}{\beta\|\overline{y}\|^2}\right)^{\frac{3-d}{4}} K_{\frac{3-d}{2}}\left(\sqrt{2\|\overline{y}\|^2/\beta}\right)d\overline{y}.$$

For $d = 2$, we get

$$\overline{\nu}_{1/2,\beta}(d\overline{y}) = \frac{\sqrt{\beta}}{4\Gamma(\beta)\sqrt{2\pi}}\left(\frac{2}{\beta\|\overline{y}\|^2}\right)^{\frac{1}{4}} K_1\left(\sqrt{2\|\overline{y}\|^2/\beta}\right)d\overline{y}.$$

2.4 Simulation and estimations

Estimation theory is a branch of statistics that deals with estimating the values of parameters based on measured empirical data that has a random component. An estimator attempts to approximate the unknown parameters using the measurements. The objective of this chapter is to estimate the parameters α and β of the normal α-stable distribution using the moment and the tailed regression methods.

The class of normal $\alpha-$stable distributions with $0 < \alpha < 1$ has been introduced by Louati *et al.* (2015) constitutes the natural extension of the normal inverse Gaussian one, given by Barndorff-Nielsen (1997). This class can be defined as the normal variance-mean mixtures where the mixing densities are the drifted $\alpha-$stable distributions. It also represents a particular case of the so-called the normal tempered stable distribution introduced by Barndorff-Nielsen and Shephard (2001). The importance of the normal α-stable distribution can be explained in financial modeling, since in this case the error is not Gaussian and the variance is random. Then, we suppose that the log returns index of prices follows a normal α-stable distribution. In fact, we assume that the conditional distribution of the log returns prices $\overline{Y}$ given the volatility $Y_1 = y_1$ is centered normal $N(0, y_1)$ and that the volatility Y_1 is a hidden random variable with drifted α-stable distribution $\pi_{\alpha,\beta}$ given in Remark 1.3.6. It follows that the mixture distribution of the log returns index of prices $\overline{Y}$ is given by

$$f_{\alpha,\beta}(\overline{y}) = \int_0^{+\infty} N(0, y_1)(\overline{y})\pi_{\alpha,\beta}(dy_1).$$

The conditional distribution of the volatility Y_1, given the log returns index of prices $\overline{Y} = \overline{y}$ is defined by

$$f_{Y_1|\overline{Y}=\overline{y}}(y_1) = \frac{N(0, y_1)(\overline{y})\pi_{\alpha,\beta}(y_1)}{f_{\alpha,\beta}(\overline{y})}$$

which represents the posteriori distribution of the volatility. Then, the conditional expectation of Y_1, given $\overline{Y} = \overline{y}$ is given by

$$\mathbb{E}(Y_1|\overline{Y} = \overline{y}) = \int_0^{+\infty} \frac{y_1 N(0, y_1)(\overline{y})\pi_{\alpha,\beta}(y_1)}{f_{\alpha,\beta}(\overline{y})} dy_1.$$

This represents the mean of the volatility, given the log returns index of prices. By the Monte-Carlo method, we can give an approximation of this expectation. The aim of this subsection is to give an algorithm allowing to simulate independent observations from the normal $\alpha-$stable model, for which the parameters are estimated using both the moment and the tailed regression methods. The asymptotic results of the obtained estimators are also supported by simulation studies.

2.4.1 Parameter's estimations

Let $F = \{P_\theta, \quad \theta \in \Theta\}$ be a parameters statistical model, where Θ is a set of unknown parameters. Let $X_1, \ldots, X_n$ be a sample with same probability distribution P_θ. An estimator $\widehat{\theta} = t(X_1, ..., X_n)$ attempts to approximate the unknown parameter θ using the sample (measurements). Since t is a function of a random sample itself is a random variable and its behavior will depend on the size of the sample, n. So, what do we look for in a good estimator? Obviously, we want our estimate to be close to the true value. Also, we want $\widehat{\theta}$ to behave in a nice way as the sample size, n, increases. If we take a large sample we would like the estimate to be more accurate than a small sample.

2.4.1.1 The method of moments

Now we recall the method of moments estimation method. The motivation behind the method of moments is that if we have a good estimator $\widehat{\theta}$, the distribution that underlies $\widehat{\theta}$ should be similar to the distribution of θ, where similarity is compared by equality of moments. However, we don't know the moments of the distribution that corresponds to θ since it is an unknown parameter. For this reason, we approximate it by the sample moment computed by the given sample

Let $\mu_1, \ldots, \mu_k$ be the first k moments of a random variable X with distribution $P_{\theta_1,\ldots,\theta_k}$. Each moment can be expressed as functions of $\theta_1, \theta_2, \ldots, \theta_k$:

$$\mu_1 \equiv E[X] = g_1(\theta_1, \theta_2, \ldots, \theta_k),$$

$$\mu_2 \equiv E[X^2] = g_2(\theta_1, \theta_2, \ldots, \theta_k),$$
$$\vdots$$
$$\mu_k \equiv E[X^k] = g_k(\theta_1, \theta_2, \ldots, \theta_k).$$

Let $X_1, \ldots, X_k$ be a sample of X with size n. For $j = 1, \ldots, k$, we denote by

$$\hat{\mu}_j = \frac{1}{n} \sum_{i=1}^{n} X_i^j$$

the jth sample moment, an estimator of μ_j. The moments estimator of $\theta_1, \theta_2, \ldots, \theta_k$ denoted by $\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_k$ and defined as a solution (if there exist) of the equations:

$$\hat{\mu}_1 = g_1(\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_k),$$
$$\hat{\mu}_2 = g_2(\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_k),$$
$$\vdots$$
$$\hat{\mu}_k = g_k(\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_k).$$

2.4.1.2 The regression method

Given a data set $\{y_i, x_{i_1}, \ldots, x_{i_p}\}_{i=1}^n$ of n statistical units, a linear regression model assumes that the relationship between the dependent variable y_i and the p-vector of regressors x_i is linear. This relationship is modeled through a disturbance term or error variable ε_i an unobserved random variable that adds noise to the linear relationship between the dependent variable and regressors. Thus the model takes the form

$$y_i = \theta_1 x_{i_1} + \cdots + \theta_p x_{i_p} + \varepsilon_i = \mathbf{x}_i^{'}\boldsymbol{\theta} + \varepsilon_i, \text{ for all } i = 1, \ldots, n,$$

where $\mathbf{x}_i^{'} = (x_{i1}, \ldots, x_{ip})$ denotes the transpose of $\mathbf{x}_i$, so that $\mathbf{x}_i^{'}\boldsymbol{\theta}$ is the inner product between vectors $\mathbf{x}_i$ and $\boldsymbol{\theta}$. Often these n equations are stacked together and written in vector form as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon},$$

where

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} \mathbf{x}_1^{'} \\ \mathbf{x}_2^{'} \\ \vdots \\ \mathbf{x}_n^{'} \end{pmatrix} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ x_{21} & \cdots & x_{2p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix}, \quad \boldsymbol{\theta} = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

The objective of the method is to estimate the parameters of the model, based on the observed pairs of values and applying a certain criterium function (the observed pairs of values are constituted by selected values of the auxiliary variable and by the corresponding observed values of the response variable). In the least squares method the estimators are the values of θ which minimize the object function. Thus, one has to calculate the derivatives, equate them to zero and solve the system of equations in $\boldsymbol{\theta}$. Suppose that $n > p$ and $rank(\mathbf{X}) = p$. Then, by applying the least squares estimation method, we deduce that the estimator $\widehat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}$ is given by the following formula

$$\widehat{\boldsymbol{\theta}} = (\mathbf{X}^{'}\mathbf{X})^{-1}\mathbf{X}^{'}\mathbf{Y}. \tag{2.28}$$

Note that, if for $i \in \{1, \ldots, n\}$, the ε_i are independent and identically distributed, then estimator is the linear unbiased estimate with the smallest variance. The linear regression also has a nice geometrical interpretation. The prediction is the orthogonal projection of the vector defined by the data to the hyper-plane defined by the regression model.

2.4.2 Simulation of the normal α-stable distribution

This part deals with the simulation of the normal α-stable model. Let us consider a random vector (Y_1, Y) on $(0, +\infty) \times \mathbb{R}$ such that Y_1 is drifted α-stable distributed with a Laplace transform $L_{\pi_{\alpha,\beta}}(s) = e^{1-(1-\beta s)^{\alpha}}$, where $\pi_{\alpha,\beta}$ is the distribution of Y_1 (see Letac and Seshadri (2005) and Louati *et al.* (2015)), and the conditional distribution of Y, given $Y_1 = z$ is a centered normal random variable with a variance $z > 0$. Hence, the distribution of Y, denoted by $\overline{\mu}_{\alpha,\beta}$, is called the normal α-stable distribution. Moreover, its density is given by

$$f_{\alpha,\beta}(y) = \int_0^{+\infty} \frac{1}{\sqrt{2\pi z}} e^{-\frac{y^2}{2z}} \pi_{\alpha,\beta}(dz).$$

By taking $d = 2$ in (2.2), we deduce that, for all $\theta \in (-\sqrt{2/\beta}, \sqrt{2/\beta})$,

$$L_{\overline{\mu}_{\alpha,\beta}}(\theta) = e^{1-\left(1-\beta\frac{\theta^2}{2}\right)^{\alpha}}. \tag{2.29}$$

In order to simulate the normal α-stable distribution, we use the Monte-Carlo approximation and the inverse transform sampling method.

In fact, consider n independent observations $z_1, z_2, \ldots, z_n$ of the drifted α-stable distribution $\pi_{\alpha,\beta}$. By using the simulation of the standard stable one given by Nolan (2007), page 20, and the Monte-Carlo simulation, we can express the normal α-stable density as follows:

$$f_{\alpha,\beta}(y) = \lim_{n\to+\infty} \frac{1}{n} \sum_{i=1}^{n} \frac{1}{\sqrt{2\pi z_i}} e^{-\frac{y^2}{2z_i}}.$$

Hence, we get Figure 2.1 which represents the curves of the normal α-stable distribution with different values of α and β.

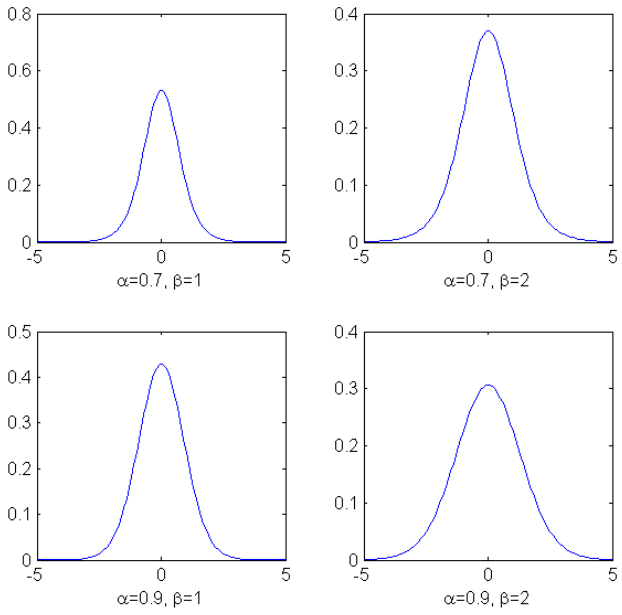


Figure 2.1: Curves of the normal $\alpha-$stable density with different values of α and β.

We simulate a sample with size $N = 1000$ from the normal α-stable distribution with different values of α and β by using the inverse transform sampling method which represents a basic method for pseudo-random number sampling (see Figure 2.2).

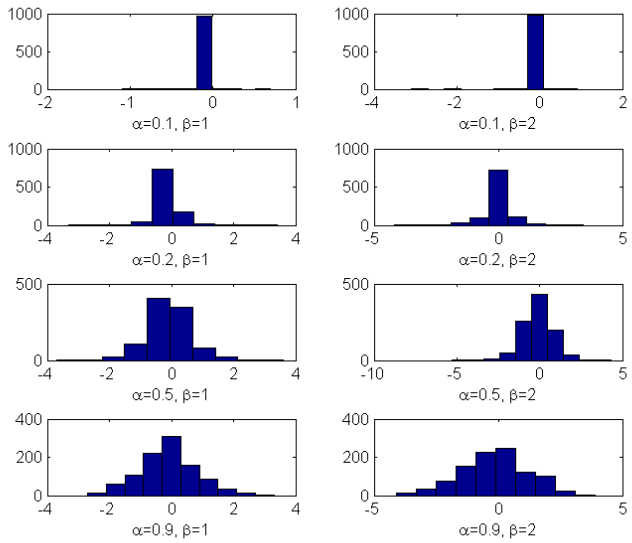


Figure 2.2: Histograms of the normal $\alpha-$stable distribution with sample size $N = 1000$.

From the histograms, we remark that the observations are more concentrated around zero, when α tends to 0, and are less concentrated around zero, when α tends to 1. This is consistent with Figure 2.2.

Notice that the variation of the parameter α gives different forms of the normal α-stable distribution. When α tends to 0, the normal α-stable converges in distribution, to the Dirac measure δ_0 (see Figure 2.1, $\alpha = 0.1$). In fact, when α tends to 0, equation (2.29) converges to $L_{\delta_0}(\theta) = 1$, and when α tends to 1, the normal α-stable converges, in distribution, to the normal one (see Figure 2.1 and Figure 2.2, $\alpha = 0.9$). This is due to the fact that the Laplace transform given by (2.29) converges, when α tends to 1, to

$$L_{N(0,1/\beta)}(\theta) = e^{\beta\frac{\theta^2}{2}}.$$

Since $f_{\alpha,\beta}$ is a symmetric distribution, then $\mathbb{E}(Y) = 0$ and $\mathbb{E}(Y^3) = 0$. This implies that the skewness is equal to 0. Using the fact that

$$\mathbb{E}(Y^2) = \frac{\partial^2 L_Y(\theta)}{\partial\theta^2}\Big|_{\theta=0} = \alpha\beta \quad \text{and} \quad \mathbb{E}(Y^4) = \frac{\partial^4 L_Y(\theta)}{\partial\theta^4}\Big|_{\theta=0} = 3\alpha\beta^2. \tag{2.30}$$

Then the kurtosis of the normal α-stable distribution, which represents a measure of the tailedness of a probability distribution, is equal to

$$\frac{\mathbb{E}(Y^4)}{\mathbb{E}(Y^2)^2} = \frac{3}{\alpha}. \tag{2.31}$$

Since $\alpha \in (0, 1)$, then the kurtosis of the normal α-stable distribution is greater than 3 (*i.e.,* the kurtosis of the normal one). It follows that the normal α-stable distribution is Leptokurtic one. So, it has a more acute peak around the mean and fatter tails that asymptotically approach zero more slowly than a Gaussian distribution.

2.4.3 Estimation of parameters using the method of moments

In this part, we estimate the parameters α and β of the normal α-stable distribution by the method of moments (MM) (see Pearson (1936)). We illustrate our work by a simulation study. In what follows, $y_1, y_2, \ldots, y_n$ are n independent observations from a normal α-stable distribution with parameters α and β.

$$\overline{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

is the empirical mean,

$$S_2 = \frac{1}{n}\sum_{i=1}^{n} y_i^2$$

is the empirical variance and

$$S_4 = \frac{1}{n}\sum_{i=1}^{n} y_i^4$$

represents the empirical centered moment with order 4.

According to (2.29), if Y is a random normal $\alpha-$stable distributed, then we get

$$\mathbb{E}(Y^2) = \frac{\partial^2 L_{\overline{\mu}_{\alpha,\beta}}}{\partial\theta^2}(0) = \alpha\beta \quad \text{and} \quad \mathbb{E}(Y^4) = \frac{\partial^4 L_{\overline{\mu}_{\alpha,\beta}}}{\partial\theta^4}(0) = 3\alpha\beta^2.$$

This allows us to give, in the following proposition, the estimator of (α, β) by using the method of moments (MM). More precisely, we have

Proposition 2.4.1 *The estimators of α and β, namely $\widehat{\alpha}$ and $\widehat{\beta}$, are given by*

$$\widehat{\alpha} = \frac{3S_2^2}{S_4} \quad \textit{and} \quad \widehat{\beta} = \frac{S_4}{3S_2}.$$

By using the Law of large numbers, we can show that the estimators $\widehat{\alpha}$ and $\widehat{\beta}$ converge, almost surely, to the true parameters α and β when n tends to $+\infty$.

We illustrate the estimations of the two parameters by generating $N = 1000$ observations from the normal α-stable distribution $\overline{\mu}_{\alpha,\beta}$. Since we have computed the estimators $\widehat{\alpha}$ and $\widehat{\beta}$ by using the MM, then we can repeat this 1000 times, hence obtaining $K = 1000$ samples of the estimator $\widehat{\alpha}$ and $\widehat{\beta}$.

The results of our simulations are summarized in Table 2.1, where we give the parameter's mean estimates their mean squared error, defined by:

$$MSE(\widehat{\alpha}) = \frac{1}{K}\sum_{i=1}^{K}(\widehat{\alpha}_i - \alpha)^2 \quad \text{and} \quad MSE(\widehat{\beta}) = \frac{1}{K}\sum_{i=1}^{K}(\widehat{\beta}_i - \beta)^2,$$

and their variation coefficients, defined as follows:

$$CV(\widehat{\alpha}) = \frac{\sqrt{MSE(\widehat{\alpha})}}{\mathbb{E}(\widehat{\alpha})} \quad \text{and} \quad CV(\widehat{\beta}) = \frac{\sqrt{MSE(\widehat{\beta})}}{\mathbb{E}(\widehat{\beta})}.$$

Moreover, by replacing the true parameters α and β by their estimators $\widehat{\alpha}$ and $\widehat{\beta}$ in the density $f_{\alpha,\beta}$ of the normal α-stable distribution, we obtain the estimated density denoted by $\widehat{f}_{\widehat{\alpha},\widehat{\beta}}$. So, we give also the mean squared error of the estimated density $\widehat{f}_{\widehat{\alpha},\widehat{\beta}}$ which is defined by

$$MSE(\widehat{f}_{\widehat{\alpha},\widehat{\beta}}) = \frac{1}{K}\sum_{i=1}^{K}(f_{\alpha,\beta}(x_i) - \widehat{f}_{\widehat{\alpha},\widehat{\beta}}(x_i))^2.$$

We compare the mean squared error for fixed true parameter α with the two values of β ($\beta = 1$ and $\beta = 2$). This table clearly shows that the mean estimates are better for fixed values of α

α	β	$\widehat{\alpha}$	$\widehat{\beta}$	$MSE\left(\widehat{\alpha}\right)$	$MSE\left(\widehat{\beta}\right)$	$CV\left(\widehat{\alpha}\right)$	$CV\left(\widehat{\beta}\right)$	$MSE\left(\widehat{f}_{\widehat{\alpha},\widehat{\beta}}\right)$
0.2	1	0.2178	0.9227	0.0029	0.0922	0.2547	0.3402	0.0154
0.2	2	0.2493	1.5286	0.0043	0.3246	0.2508	0.3805	0.0055
0.5	1	0.5402	0.9444	0.0075	0.0348	0.1622	0.1871	8.2892×10^{-5}
0.5	2	0.6049	1.6033	0.0142	0.1842	0.1944	0.2738	1.3472×10^{-5}
0.7	1	0.7364	0.9621	0.0077	0.0186	0.1179	0.1471	1.9652×10^{-6}
0.7	2	0.8022	1.7052	0.0132	0.1117	0.1400	0.1957	4.5169×10^{-6}

Table 2.1: Statistical study of the estimators, their MSEs, as well as their CVs

with $\beta = 1$ than for $\beta = 2$. We also remark that the mean squared error is lower for $\beta = 1$ than for $\beta = 2$. Moreover, we have the same remark if we compare the variation coefficient of $\widehat{\alpha}$ and $\widehat{\beta}$. Likewise, the mean squared error of the estimated density $\widehat{f}_{\widehat{\alpha},\widehat{\beta}}$ is small when $\alpha = 0.2$ and it is even smaller when $\alpha = 0.5$ or 0.7. Hence, we may deduce that the more α increases, the more $\widehat{f}_{\widehat{\alpha},\widehat{\beta}}$ gets closer and closer to the true density $f_{\alpha,\beta}$.

Now, we are ready to make a simulation study of the estimators $\widehat{\alpha}$ and $\widehat{\beta}$ of α and β. By using the MM, we have simulated $K = 1000$ realizations of $\widehat{\alpha}$ and $\widehat{\beta}$. So, Figure 2.3 and Figure 2.4 represent the histograms of the estimators $\widehat{\alpha}$ and $\widehat{\beta}$.

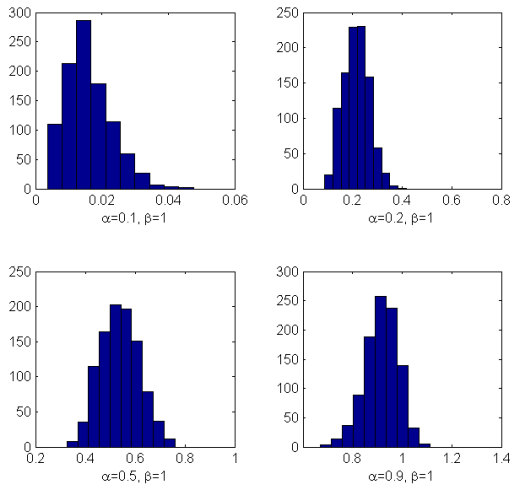


Figure 2.3: Histograms of the estimator $\widehat{\alpha}$ by the MM for $K = 1000$ samples.

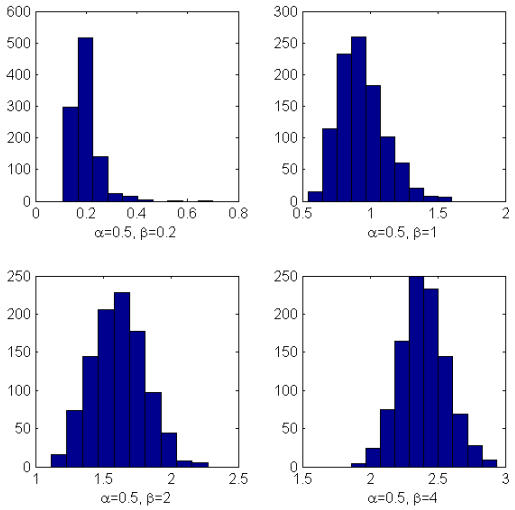


Figure 2.4: Histograms of the estimator $\widehat{\beta}$ by the MM for $K = 1000$ samples.

Note that the histograms of the estimator $\widehat{\alpha}$ have the form of the normal distribution with a mean α. Furthermore, the histograms of the estimator $\widehat{\beta}$ can be approximated by a normal distribution for large values of β (see Figure 2.4).

In order to test the performance of our estimators $\widehat{\alpha}$ and $\widehat{\beta}$, we use the mean squared error. Figure 2.5 and 2.6 represent the curves of the $MSE(\widehat{\alpha})$ and $MSE(\widehat{\beta})$ for $\alpha \in \{0.2, 0.5\}$ and for $\beta = 1$ by the MM for K $= 1000$ samples.

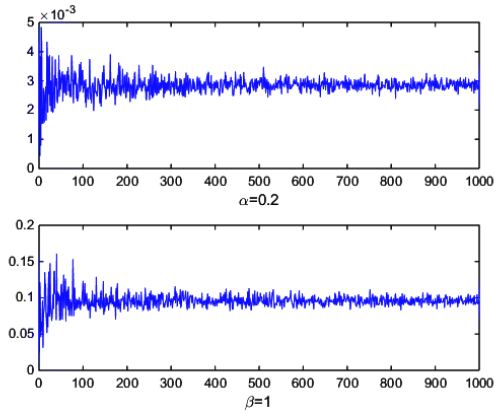


Figure 2.5: Curves of the $MSE(\widehat{\alpha})$ and $MSE(\widehat{\beta})$ by the MM when $\alpha = 0.2$, $\beta = 1$.

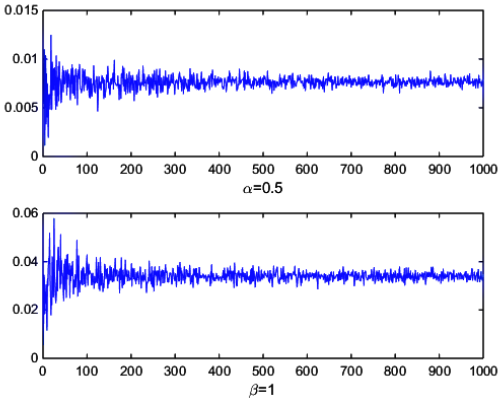


Figure 2.6: Curves of the $MSE(\widehat{\alpha})$ and $MSE(\widehat{\beta})$ by the MM when $\alpha = 0.5$, $\beta = 1$.

In order to compare the true and the estimated densities $f_{\alpha,\beta}$ and $\widehat{f}_{\widehat{\alpha},\widehat{\beta}}$, we have Figure 2.7 which represents the curves of $f_{\alpha,\beta}$ and $\widehat{f}_{\widehat{\alpha},\widehat{\beta}}$ for different values of α and β.

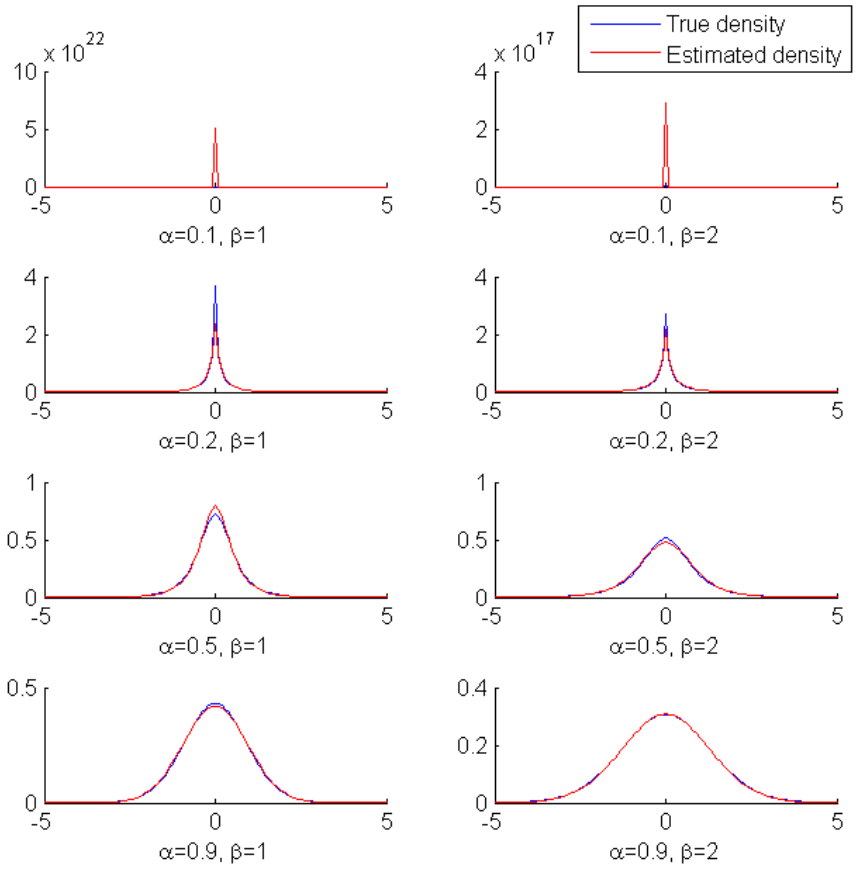


Figure 2.7: Curves of the densities $f_{\alpha,\beta}$ and $\widehat{f}$ by the MM.

It is important to notice that the curves of $f_{\alpha,\beta}$ and $\widehat{f}_{\widehat{\alpha},\widehat{\beta}}$ are similar when $\alpha = 0.2,\ 0.5$ and 0.7 and they are different for $\alpha = 0.1$. In fact, the mean squared error $MSE\left(\widehat{f}_{\widehat{\alpha},\widehat{\beta}}\right)$ is higher when α tends to 0. This explains the difference between the two densities when $\alpha = 0.1$.

Figure 2.8 represents the $MSE(\widehat{f}_{\widehat{\alpha},\widehat{\beta}})$ for $\alpha = 0.5$ and $\beta = 0.7$ of the estimated $\widehat{\alpha}$ and $\widehat{\beta}$ by using the MM.

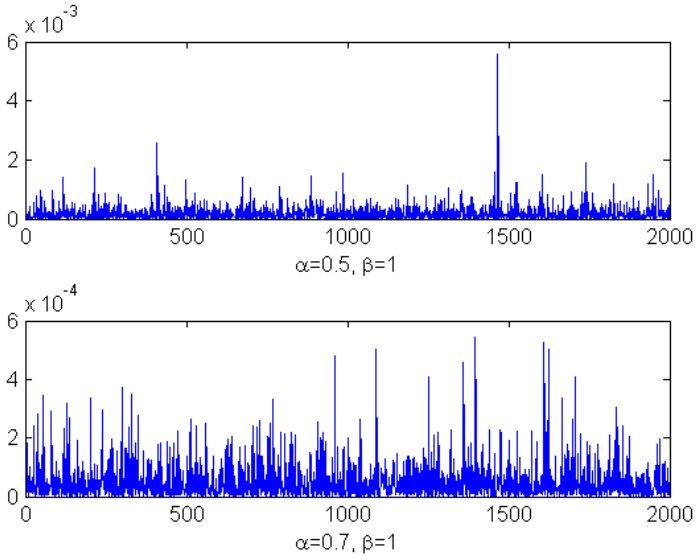


Figure 2.8: Curves of the $MSE(\widehat{f})$ for $\alpha = 0.5$ and $\alpha = 0.7$.

The mean squared error between the true and the estimated density are less than 6×10^{-3} for $\alpha = 0.2$ and $\beta = 1$ and less than 6×10^{-4} for $\alpha = 0.7$ and $\beta = 1$. This gives an idea about the similarity between the density $f_{\alpha,\beta}$ and its estimation $\widehat{f}_{\widehat{\alpha},\widehat{\beta}}$.

2.4.4 Estimation of parameters using the tailed regression method

In this section, we will present a new estimation method, which we have called the tailed regression method (TRM). An important result, which is due to Barndorff-Nielsen and Shephard (2001), says that when y tends to $+\infty$, the probability density function $f_{\alpha,\beta}(y)$ of the normal α-stable distribution $\overline{\mu}_{\alpha,\beta}$, is equivalent to

$$f_{\alpha,\beta}(y) \sim c_{\alpha,\beta} y^{-\alpha-1} e^{-\sqrt{\frac{2}{\beta}}y},$$

where

$$c_{\alpha,\beta} = 2^{\alpha+1} \left(\beta/2\right)^{\frac{\alpha}{2}-\frac{1}{4}} e^{1} \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)\Gamma(1-\alpha)}.$$

In order to introduce the estimation of α and β by using the TRM, we need to get an approximation of the cumulative function $F(y) = \int_{-\infty}^{y} f_{\alpha,\beta}(x)dx$ of the normal α-stable distribution $\overline{\mu}_{\alpha,\beta}$. In order to get the approximation, we need this classical technical lemma.

Lemma 2.4.2 *Let g and h be two positive continuous functions on $[a, b)$ ($b \in \overline{\mathbb{R}}$ and $a < b$) such that $g(t) \sim h(t)$, when $t \to b$. Then*

$$\int_x^b g(t)dt \sim \int_x^b h(t)dt, \text{ when } x \to b.$$

Now, we are ready to get the approximation of the cumulant function F of $\overline{\mu}_{\alpha,\beta}$.

Proposition 2.4.3 *Let Y be a random variable with normal α-stable distribution $\overline{\mu}_{\alpha,\beta}$. When A tends to $+\infty$, we have the following approximation*

$$F(A) = \mathbb{P}(Y \leq A) \sim 1 - c_{\alpha,\beta} A^{-\alpha-1} \sqrt{\frac{\beta}{2}} e^{-\sqrt{\frac{2}{\beta}}A}. \tag{2.32}$$

Proof

In order to determine an equivalent of $\overline{F}(A) = 1 - F(A)$, when $A \to +\infty$, we need first to determine an equivalent of $f_{\alpha,\beta}(y)$, when $y \to +\infty$. Then according to Barndorff-Nielsen and Shephard (2001), page 7, we have

$$f_{\alpha,\beta}(y) = \frac{1}{\sqrt{2\pi}} \int_0^{+\infty} \frac{e^{-\frac{y^2}{2z}}}{\sqrt{z}} \pi_{\alpha,\beta}(dz) \sim c_{\alpha,\beta} y^{-\alpha-1} e^{-\sqrt{\frac{2}{\beta}}y},$$

where $c_{\alpha,\beta} = 2^{\alpha+1} (\beta/2)^{\frac{\alpha}{2}-\frac{1}{4}} e^1 \dfrac{\Gamma(\alpha+1)}{\Gamma(\alpha)\Gamma(1-\alpha)}$.
This equivalence relation implies that

$$\overline{F}(A) = \int_A^{+\infty} \overline{\mu}_{\alpha,\beta}(dy) \sim \int_A^{+\infty} c_{\alpha,\beta} y^{-\alpha-1} e^{-\sqrt{\frac{2}{\beta}}y} dy, \quad \text{when} \quad A \to +\infty. \tag{2.33}$$

So, our aim now is to determine an equivalent of

$$\int_A^{+\infty} c_{\alpha,\beta} y^{-\alpha-1} e^{-\sqrt{\frac{2}{\beta}}y} dy, \text{ when } A \to +\infty.$$

Hence, a two integrations by part imply

$$\begin{aligned}
&\int_A^{+\infty} c_{\alpha,\beta} y^{-\alpha-1} e^{-\sqrt{\frac{2}{\beta}}y} dy \\
&= c_{\alpha,\beta} A^{-\alpha-1}\sqrt{\frac{\beta}{2}} e^{-\sqrt{\frac{2}{\beta}}A} - c_{\alpha,\beta}(\alpha+1)\sqrt{\frac{\beta}{2}}\int_A^{+\infty} y^{-\alpha-2} e^{-\sqrt{\frac{2}{\beta}}y} dy \\
&= c_{\alpha,\beta} A^{-\alpha-1}\sqrt{\frac{\beta}{2}} e^{-\sqrt{\frac{2}{\beta}}A} - c_{\alpha,\beta}(\alpha+1)\frac{\beta}{2}\left(A^{-\alpha-2} e^{-\sqrt{\frac{2}{\beta}}A} - (\alpha+2)\int_A^{+\infty} y^{-\alpha-3} e^{-\sqrt{\frac{2}{\beta}}y} dy\right) \\
&= c_{\alpha,\beta} A^{-\alpha-1}\sqrt{\frac{\beta}{2}} e^{-\sqrt{\frac{2}{\beta}}A} - c_{\alpha,\beta}(\alpha+1) A^{-\alpha}\frac{\beta}{2} e^{-\sqrt{\frac{2}{\beta}}A} K(A),
\end{aligned}$$

where

$$K(A) = A^{-2} - (\alpha+2) A^{\alpha} e^{\sqrt{\frac{2}{\beta}}A}\int_A^{+\infty} y^{-\alpha-3} e^{-\sqrt{\frac{2}{\beta}}y} dy.$$

Since

$$0 \leq \int_A^{+\infty} y^{-\alpha-3} e^{-\sqrt{\frac{2}{\beta}}y} dy \leq A^{-\alpha-3}\int_A^{+\infty} e^{-\sqrt{\frac{2}{\beta}}y} dy = A^{-\alpha-3}\sqrt{\frac{\beta}{2}} e^{-\sqrt{\frac{2}{\beta}}A},$$

then, we get

$$A^{-2} - A^{-3}(\alpha+2)\sqrt{\frac{\beta}{2}} \leq K(A) \leq A^{-2}.$$

This implies that

$$K(A) \longrightarrow 0, \text{ when } A \to +\infty.$$

It follows that

$$\int_A^{+\infty} c_{\alpha,\beta} y^{-\alpha-1} e^{-\sqrt{\frac{2}{\beta}}y} dy \sim c_{\alpha,\beta} A^{-\alpha-1}\sqrt{\frac{\beta}{2}} e^{-\sqrt{\frac{2}{\beta}}A}, \quad \textit{when} \quad A \to +\infty.$$

Using (2.33), we get

$$\overline{F}(A) = \mathbb{P}(Y > A) = \int_A^{+\infty} \overline{\mu}_{\alpha,\beta}(dy) \sim c_{\alpha,\beta} A^{-\alpha-1}\sqrt{\frac{\beta}{2}} e^{-\sqrt{\frac{2}{\beta}}A}, \quad \textit{when} \quad A \to +\infty.$$

Hence, we get the desired result. □

From this proposition, we deduce that there exists a function $\epsilon : \mathbb{R} \longrightarrow \mathbb{R}$ satisfies the condition $\epsilon(A) \longrightarrow 0$, when $A \longrightarrow +\infty$, such that

$$\overline{F}(A) = \mathbb{P}(Y > A) = c_{\alpha,\beta} A^{-\alpha-1}\sqrt{\frac{\beta}{2}} e^{-\sqrt{\frac{2}{\beta}}A}(1 + \epsilon(A)).$$

This implies that

$$\ln(\overline{F}(A)) = \ln\left(c_{\alpha,\beta}\sqrt{\frac{\beta}{2}}\right) - \sqrt{\frac{2}{\beta}}A - (\alpha+1)\ln(A) + \ln(1+\epsilon(A)). \tag{2.34}$$

Now let us consider $Y_1, Y_2, \ldots, Y_n$ be n copies of the normal α-stable distribution $\overline{\mu}_{\alpha,\beta}$ and let $a_1, a_2, \ldots, a_n$ be n points on the tail of the empirical cumulative function distribution such that $F_{Y_1}^{-1}(0.8 + 0.19\frac{i-1}{n}) = a_i$, $i \in \{1, \ldots, n\}$. According to the Law of large numbers, for n large enough, we have the following approximation

$$y_i = \frac{1}{n}\sum_{j=1}^{n} \mathbf{1}_{\{Y_j > a_i\}} \approx \mathbb{P}(Y > a_i).$$

According to (2.34), and for all $i \in \{1, 2, \ldots, n\}$, we get

$$\ln(y_i) = \ln\left(c_{\alpha,\beta}\sqrt{\frac{\beta}{2}}\right) - \sqrt{\frac{2}{\beta}}a_i - (\alpha+1)\ln(a_i) + \epsilon_i,$$

where ϵ_i is the error random variable.

Then, the empirical mean of $(\ln(y_i))_{1\leq i\leq n}$ is given by

$$\overline{\ln(y)} = \ln\left(c_{\alpha,\beta}\sqrt{\frac{\beta}{2}}\right) - \sqrt{\frac{2}{\beta}}\overline{a} - (\alpha+1)\overline{\ln(a)} + \overline{\epsilon},$$

where $\overline{\ln(y)} = \frac{1}{n}\sum_{i=1}^{n}\ln(y_i)$, $\overline{a} = \frac{1}{n}\sum_{i=1}^{n} a_i$, $\overline{\ln(a)} = \frac{1}{n}\sum_{i=1}^{n}\ln(a_i)$ and $\overline{\epsilon} = \frac{1}{n}\sum_{i=1}^{n}\epsilon_i$.

Therefore,

$$\ln(y_i) - \overline{\ln(y)} = -\sqrt{\frac{2}{\beta}}(a_i - \overline{a}) - (\alpha+1)(\ln(a_i) - \overline{\ln(a)}) + (\epsilon_i - \overline{\epsilon}), \text{ for all } i \in \{1, 2, \ldots, n\}. \tag{2.35}$$

By taking $\mathbf{X} = \begin{pmatrix} \ln(a_1) - \overline{\ln(a)} & a_1 - \overline{a} \\ \ln(a_2) - \overline{\ln(a)} & a_2 - \overline{a} \\ \vdots & \vdots \\ \ln(a_n) - \overline{\ln(a)} & a_n - \overline{a} \end{pmatrix}$, $\mathbf{Y} = \begin{pmatrix} \ln y_1 - \overline{\ln y} \\ \vdots \\ \ln y_n - \overline{\ln y} \end{pmatrix}$, $\boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 - \overline{\epsilon} \\ \vdots \\ \epsilon_n - \overline{\epsilon} \end{pmatrix}$,

$\theta = \begin{pmatrix} -\alpha - 1 \\ -\sqrt{2/\beta} \end{pmatrix}$ and by using (2.35), we can write

$$\mathbf{Y} = \mathbf{X}\theta + \boldsymbol{\epsilon}.$$

Then, according to (2.28), we deduce that the estimator $\widehat{\theta}$ of θ is given by

$$\widehat{\theta} = (\mathbf{X}^{'}\mathbf{X})^{-1}\mathbf{X}^{'}\mathbf{Y}.$$

Let us notice that $\widehat{\theta}$ is an unbiased estimator of θ.

Next, we estimate the parameters α and β by using the tailed regression method.

Proposition 2.4.4 *The tailed regression estimator $(\widehat{\alpha}, \widehat{\beta})$ of (α, β) is given by*

$$\widehat{\alpha} = -\langle \widehat{\theta}, e_1 \rangle - 1, \text{ and } \widehat{\beta} = \frac{2}{\langle \widehat{\theta}, e_2 \rangle^2},$$

where $\{e_1, e_2\}$ is the the standard basis of $\mathbb{R}^2$.

Note that, since n tends to $+\infty$, the obtained estimators $\widehat{\alpha}$ and $\widehat{\beta}$ converge, almost surely, and since knowing that $\widehat{\theta}$ is an unbiased estimator of θ, then $\widehat{\alpha}$ is also an unbiased estimator of α.

We illustrate our estimation of the parameters by generating $N = 1000$ observations from the normal α-stable distribution $\overline{\mu}_{\alpha,\beta}$. If we determine the estimator $\widehat{\theta}$, then we get the values of the tailed regression estimators $\widehat{\alpha}$ and $\widehat{\beta}$. We repeat this 1000 times and we get $K = 1000$ samples of the estimators $\widehat{\alpha}$ and $\widehat{\beta}$. The results of our simulations are shown in Table 2.2.

α	β	$\widehat{\alpha}$	$\widehat{\beta}$	$MSE\,(\widehat{\alpha})$	$MSE\left(\widehat{\beta}\right)$	$CV\,(\widehat{\alpha})$	$CV\left(\widehat{\beta}\right)$	$MSE\left(\widehat{f}_{\widehat{\alpha},\widehat{\beta}}\right)$
0.2	1	0.1875	1.0699	0.0016	0.0049	0.2147	0.0653	0.0047
0.2	2	0.1468	2.1259	0.0041	0.0162	0.4412	0.0598	0.4239
0.5	1	0.4951	1.0745	0.0010	0.0056	0.0630	0.0698	1.9606×10^{-5}
0.5	2	0.4626	2.1551	0.0022	0.0240	0.1031	0.0719	5.0468×10^{-5}
0.7	1	0.6876	1.0340	0.0015	0.0012	0.0534	0.0336	2.1427×10^{-4}
0.7	2	0.6707	2.0912	0.0018	0.0086	0.0638	0.0440	1.4122×10^{-4}

Table 2.2: Statistical study of the estimators, their MSEs, as well as their CVs

For fixed values of α, the MSEs of α and β are lower when $\beta = 1$ than when $\beta = 2$.

Using the TRM, we have simulated $K = 1000$ observations of $\widehat{\alpha}$ and $\widehat{\beta}$. Figure 2.9 and Figure 2.10 represent the histograms of the estimators $\widehat{\alpha}$ and $\widehat{\beta}$.

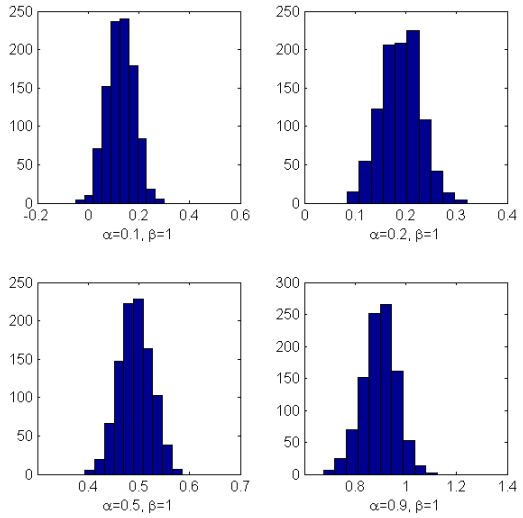


Figure 2.9: Histograms of the estimator $\widehat{\alpha}$ by the TRM for $K = 1000$ samples.

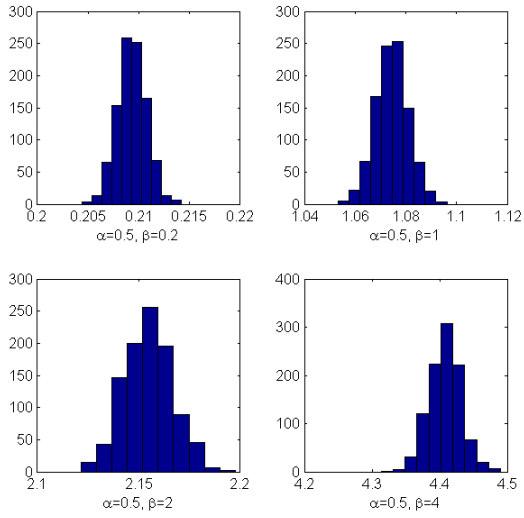


Figure 2.10: Histograms of the estimator $\widehat{\beta}$ by the TRM for $K = 1000$ samples.

Note that the histograms of the estimator $\widehat{\alpha}$ have the form of the normal distribution. Moreover, the histograms of the estimator $\widehat{\beta}$ can be also seen as a normal distribution.

In order to test the performance of our estimators $\widehat{\alpha}$ and $\widehat{\beta}$, we use the mean squared error. Figures 2.11 and 2.12 represent the curves of the MSEs of $\widehat{\alpha}$ and $\widehat{\beta}$ when $\alpha = 0.2$, $\beta = 1$ and when $\alpha = 0.5$, $\beta = 1$.

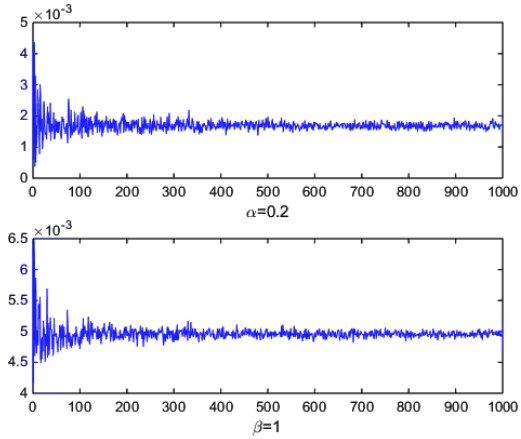


Figure 2.11: Curves of the $MSE(\widehat{\alpha})$ and $MSE(\widehat{\beta})$ by the TRM when $\alpha = 0.2$, $\beta = 1$.

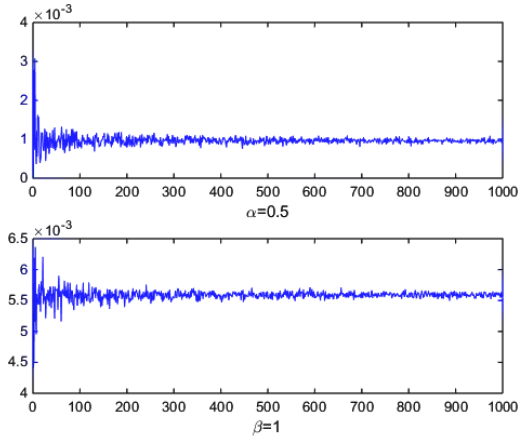


Figure 2.12: Curves of the $MSE(\widehat{\alpha})$ and $MSE(\widehat{\beta})$ by the TRM when $\alpha = 0.5$, $\beta = 1$.

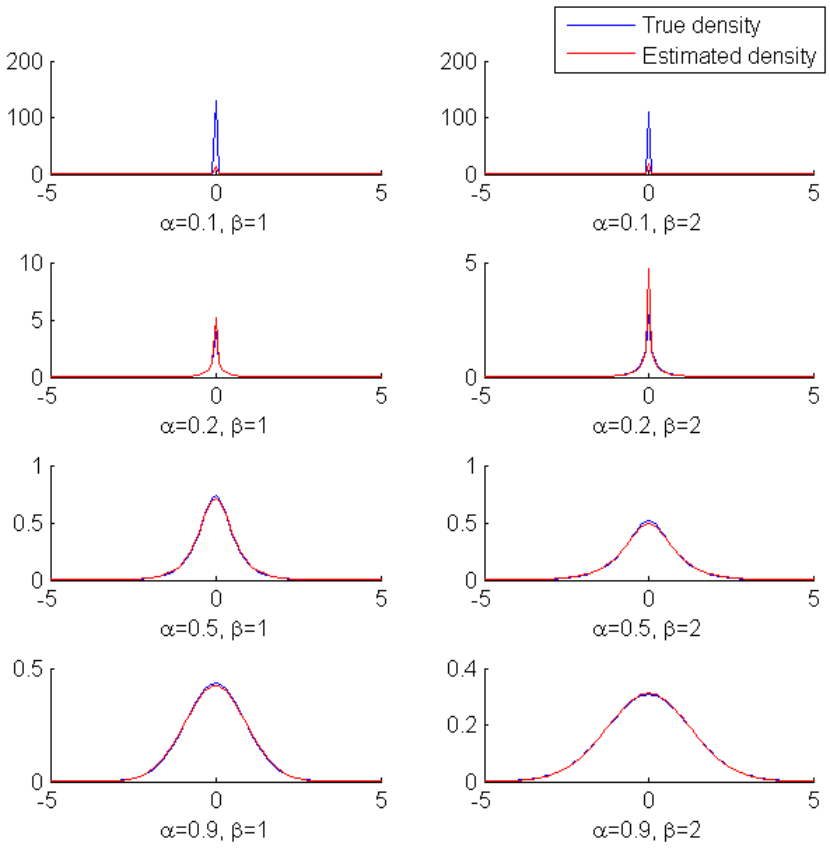


Figure 2.13: Curves of the densities $f_{\alpha,\beta}$ and $\widehat{f}$ by the TRM.

Using the estimators $\widehat{\alpha}$ and $\widehat{\beta}$, obtained by the TRM, we can get the estimated density $\widehat{f}_{\widehat{\alpha},\widehat{\beta}}$ of the normal α-stable density $f_{\alpha,\beta}$. Figure 2.13 allows us to show that the estimated and the true have very close curves.

Similarly, Figure 2.14 confirms the same finding by giving the representation of the $MSE(\widehat{f}_{\widehat{\alpha},\widehat{\beta}})$ for $\alpha = 0.5,\ \beta = 1$ and $\alpha = 0.7,\ \beta = 1$ of the estimated $\widehat{\alpha}$ and $\widehat{\beta}$ by the tailed regression method.

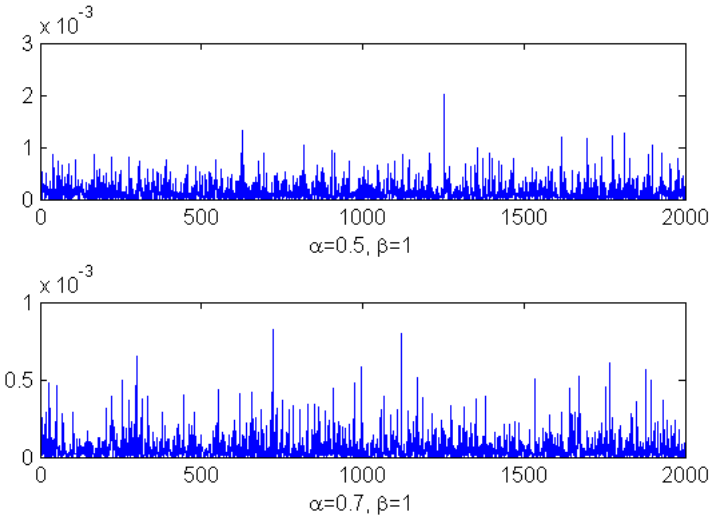


Figure 2.14: Curves of the $MSE(\widehat{f})$ for $\alpha = 0.5$ and $\alpha = 0.7$

This figure shows that the $MSE(\widehat{f}_{\widehat{\alpha},\widehat{\beta}})$ is lower than 2×10^{-3}, when $\alpha = 0.5$ and $\beta = 1$ and is lower than 0.8×10^{-3}, when $\alpha = 0.7$ and $\beta = 1$

2.4.5 Comparison

Here, we propose an approximation of the normal $\alpha-$stable distribution using the Monte-Carlo method. We get the curves of the normal $\alpha-$stable density and its cumulative function for different values of $0 < \alpha < 1$ and β. Moreover, we obtain the histograms of this distribution by using the inverse transform sampling method. Then, we estimate the parameters α and β according to two methods. More precisely, we use the MM and the tailed regression method. We illustrate our estimators through histograms for 1000 samples and we test the performance of the estimators through the mean squared error and the variation coefficient for each method. Moreover, we get the curves of the MSE of the estimators $\widehat{\alpha}$, $\widehat{\beta}$ and the curves of the estimated and true densities. These results are summarized in the following table.

α	β	$CV\,(\widehat{\alpha})\ \ (MM)$	$CV\left(\widehat{\beta}\right)\ \ (MM)$	$CV\,(\widehat{\alpha})\ \ (TRM)$	$CV\left(\widehat{\beta}\right)\ \ (TRM)$
0.2	1	0.2547	0.3402	0.2147	0.0653
0.2	2	0.2508	0.3805	0.4412	0.0598
0.5	1	0.1622	0.1871	0.0630	0.0698
0.5	2	0.1944	0.2738	0.1031	0.0719
0.7	1	0.1179	0.1471	0.0534	0.0336
0.7	2	0.1400	0.1957	0.0638	0.0440

Table 2.3: The variation coefficient for different values of α and β.

It is worth mentioning that the variation coefficient of $\widehat{\alpha}$ and $\widehat{\beta}$ are better in the TRM than in the moment one.

Chapter 3

Characterizations of multivariate stable processes.

This chapter deals with the characterization of a multivariate stochastic stable process $(Y(t))_{t\geq 0}$ using an independence property between a standard exponential random variable T and $Y(T)$. Under some independence conditions, we establish a characterization of the stochastic process $(\widetilde{Y}(t))_{t\geq 0} = (Y_1(t), \overline{Y}(t))_{t\geq 0}$ on $(0,+\infty)\times\mathbb{R}^{d-1}$, with $d\geq 2$, based on the cut property. More precisely, the distribution of $\overline{Y}(t)$ is defined as a normal variance-mean mixture with a stable mixing distribution. In this context, we study the conditional distribution of $\overline{Y}(T)$ given $Y_1(T)$ and we determine the distribution of $\widetilde{Y}(T)$. These results extend the one dimensional case.

3.1 Characterization of stable processes on $\mathbb{R}^d$

This section is devoted to the characterization of the stable processes on $\mathbb{R}^d$. To this end, we introduce the following notation. A stochastic process $Y = (Y(t))_{t\geq 0}$ and a positive random variable T are pair-wise independent if for all $t\geq 0$, $Y(t)$ and T are independent (*i.e.*, $\forall\ t\geq 0, Y(t)\perp T$). We study the case $\alpha\neq 1$ and $\alpha = 1$ separately. In the following theorem, we prove that a necessary and sufficient condition for a Lévy process to be $\alpha-$stable, is that it satisfies some independence property with a positive random variable T. For this reason, consider $Y = (Y(t))_{t\geq 0}$ as a Lévy process on $\mathbb{R}^d$ and, for all $t>0$, we denote the distribution of $Y(t)$ by μ_t. In what follows, we suppose that $\Theta(\mu_1) = \Theta(\mu_t)$ is non empty set.

Theorem 3.1.1 *Let $d \geq 1$ and let α be an element of $(0,2] \setminus \{1\}$, and $Y = (Y(t))_{t\geq 0}$ be a Lévy process on $\mathbb{R}^d$ pair-wise independent of a positive random variable T with density $f_T(t)$. Then, Y is a strictly $\alpha-$stable process if, and only if, $T^{-\frac{1}{\alpha}}Y(T) \perp T$.*

Proof

($\Longrightarrow$) Since $Y = (Y(t))_{t\geq 0}$ is a strictly stable process on $\mathbb{R}^d$, then $(Y(t))_{t\geq 0}$ is self-similar with index $1/\alpha$ (see Bertoin (1996), page 216). This implies that $t^{-\frac{1}{\alpha}}Y(t) =^{\mathcal{L}} Y(1)$. On the one hand, since T and $Y(1)$ are independent, then we get, for all $s \leq 0$ and for all $\theta \in \Theta(\mu_1)$,

$$\begin{aligned}
\mathbb{E}\left(e^{sT+\left\langle T^{-\frac{1}{\alpha}}Y(T),\theta\right\rangle}\right) &= \int_0^{+\infty} \mathbb{E}\left(e^{st+\left\langle t^{-\frac{1}{\alpha}}Y(t),\theta\right\rangle}\Big|T=t\right) f_T(t)dt \\
&= \int_0^{+\infty} e^{st}\mathbb{E}\left(e^{\langle Y(1),\theta\rangle}\right) f_T(t)dt \\
&= \mathbb{E}\left(e^{\langle Y(1),\theta\rangle}\right)\mathbb{E}\left(e^{sT}\right). \qquad (3.1)
\end{aligned}$$

On the other hand, for all $\theta \in \Theta(\mu_1)$, we have

$$\begin{aligned}
\mathbb{E}\left(e^{\left\langle T^{-\frac{1}{\alpha}}Y(T),\theta\right\rangle}\right) &= \int_0^{+\infty} \mathbb{E}\left(e^{\left\langle t^{-\frac{1}{\alpha}}Y(t),\theta\right\rangle}\Big|T=t\right) f_T(t)dt \\
&= \int_0^{+\infty} \mathbb{E}\left(e^{\langle Y(1),\theta\rangle}\right) f_T(t)dt \\
&= \mathbb{E}\left(e^{\langle Y(1),\theta\rangle}\right).
\end{aligned}$$

Inserting this in (3.1), we deduce that $T^{-\frac{1}{\alpha}}Y(T)$ and T are independent.

($\Longleftarrow$) As $T^{-\frac{1}{\alpha}}Y(T)$ and T are independent then, for all $s \leq 0$, we have

$$\mathbb{E}\left(e^{sT+\left\langle T^{-\frac{1}{\alpha}}Y(T),\theta\right\rangle}\right) = \mathbb{E}\left(e^{sT}\right)\mathbb{E}\left(e^{\left\langle T^{-\frac{1}{\alpha}}Y(T),\theta\right\rangle}\right).$$

Hence, we obtain, for all $s \leq 0$,

$$\int_0^{+\infty} \mathbb{E}\left(e^{st+\left\langle t^{-\frac{1}{\alpha}}Y(t),\theta\right\rangle}\Big|T=t\right) f_T(t)dt = \mathbb{E}\left(e^{sT}\right)\int_0^{+\infty} \mathbb{E}\left(e^{\left\langle t^{-\frac{1}{\alpha}}Y(t),\theta\right\rangle}\Big|T=t\right) f_T(t)dt.$$

This implies that, for all $\theta \in \mathbb{R}^d$ such that $t^{-\frac{1}{\alpha}}\theta \in \Theta(\mu_t)$,

$$\begin{aligned}\int_0^{+\infty} e^{st} L_{\mu_t}(t^{-\frac{1}{\alpha}}\theta) f_T(t)dt &= \int_0^{+\infty} e^{st}\mathbb{E}\left(e^{\left\langle Y(t), t^{-\frac{1}{\alpha}}\theta\right\rangle}\right) f_T(t)dt \\ &= \mathbb{E}\left(e^{sT}\right)\int_0^{+\infty}\mathbb{E}\left(e^{\left\langle Y(t), t^{-\frac{1}{\alpha}}\theta\right\rangle}\right) f_T(t)dt \\ &= \mathbb{E}\left(e^{sT}\right)\int_0^{+\infty} L_{\mu_t}(t^{-\frac{1}{\alpha}}\theta) f_T(t)dt.\end{aligned}$$

Since μ_t is infinitely divisible then, for all $\theta \in \Theta(\mu_1) = \Theta(\mu_t)$, we have

$$L_{\mu_t}(\theta) = \mathbb{E}(e^{\langle Y(t),\theta\rangle}) = e^{tk_{\mu_1}(\theta)}.$$

This implies that

$$\int_0^{+\infty} e^{st} e^{tk_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right)} f_T(t)dt = \mathbb{E}\left(e^{sT}\right)\int_0^{+\infty} e^{tk_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right)} f_T(t)dt. \tag{3.2}$$

Let us consider the measure

$$\mu_{\theta,\alpha}(dt) = e^{tk_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right)} f_T(t)1_{(0,+\infty)}(t)dt. \tag{3.3}$$

Combined with (3.2), we obtain, for all $s \leq 0$,

$$L_{\mu_{\theta,\alpha}}(s) = \int_0^{+\infty} e^{st}\mu_{\theta,\alpha}(dt) = \mathbb{E}\left(e^{sT}\right)\mu_{\theta,\alpha}((0,+\infty)).$$

This implies that

$$\frac{\mu_{\theta,\alpha}(dt)}{\mu_{\theta,\alpha}((0,+\infty))} = f_T(t)1_{(0,+\infty)}dt.$$

Hence, according to (3.3), we get

$$\frac{e^{tk_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right)} f_T(t)1_{(0,+\infty)}(t)dt}{\mu_{\theta,\alpha}((0,+\infty))} = f_T(t)1_{(0,+\infty)}dt.$$

Using (3.3), we deduce that the function $t \longmapsto e^{tk_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right)}$ is constant, that is for all $t > 0$, we have

$$e^{tk_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right)} = \mu_{\theta,\alpha}((0,+\infty)).$$

Differentiating this with respect to t, we deduce that

$$0 = \frac{\partial}{\partial t}\left(tk_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right)\right) = k_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right) - \frac{1}{\alpha}\left\langle k'_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right), t^{-\frac{1}{\alpha}}\theta\right\rangle.$$

Therefore, for all $t > 0$,

$$k_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right) = \frac{1}{\alpha}\left\langle k'_{\mu_1}\left(t^{-\frac{1}{\alpha}}\theta\right), t^{-\frac{1}{\alpha}}\theta\right\rangle.$$

Hence, k_{μ_1} is $\alpha-$homogeneous. According to (2.14), we deduce that the random vector $Y(1)$ is strictly $\alpha-$stable with $\alpha \neq 1$. □

Next, we turn our attention on the case $\alpha = 1$. More precisely, we have

Theorem 3.1.2 *Let $d \geq 1$ and let $Y = (Y(t))_{t\geq 0}$ be a Lévy process on $\mathbb{R}^d$ pair-wise independent of a positive random variable T with density $f_T(t)$. Then, Y is 1-stable if, and only if, there exists $m_0 \in \mathbb{R}^d \setminus \{0\}$ such that $T^{-1}Y(T) + m_0 \ln(T) \perp T$.*

Proof

($\Longrightarrow$) Since $Y = (Y(t))_{t\geq 0} \perp T$ then, for all $\theta \in \Theta(\mu_1)$ such that $t^{-1}\theta \in \Theta(\mu_1)$, we have

$$\begin{aligned}\mathbb{E}\left(e^{\langle T^{-1}Y(T)+m_0\ln(T),\theta\rangle}\right) &= \int_0^{+\infty}\mathbb{E}\left(e^{\langle t^{-1}Y(t)+m_0\ln(t),\theta\rangle}\Big|T=t\right)f_T(t)dt\\ &= \int_0^{+\infty}\mathbb{E}\left(e^{\langle Y(t),t^{-1}\theta\rangle}\right)e^{\ln(t)\langle m_0,\theta\rangle}f_T(t)dt.\end{aligned}$$

It follows that

$$\mathbb{E}\left(e^{\langle T^{-1}Y(T)+m_0\ln(T),\theta\rangle}\right) = \int_0^{+\infty} e^{tk_{\mu_1}(t^{-1}\theta)}e^{\ln(t)\langle m_0,\theta\rangle}f_T(t)dt. \tag{3.4}$$

Since $\alpha = 1$ then, using (1.7), we get

$$\begin{aligned}tk_{\mu_1}(t^{-1}\theta) &= \frac{2t}{\pi}\int_{\mathbb{S}}\langle t^{-1}\theta,s\rangle\ln|\langle t^{-1}\theta,s\rangle|\rho(ds)\\ &= \frac{2}{\pi}\int_{\mathbb{S}}\langle\theta,s\rangle\ln|\langle\theta,s\rangle|\rho(ds) - \ln(t)\frac{2}{\pi}\int_{\mathbb{S}}\langle\theta,s\rangle\rho(ds).\end{aligned}$$

Combined with (1.6), we obtain

$$tk_{\mu_1}(t^{-1}\theta) = \frac{2}{\pi}\int_{\mathbb{S}}\langle\theta,s\rangle\ln|\langle\theta,s\rangle|\rho(ds) - \ln(t)\langle m_0,\theta\rangle. \tag{3.5}$$

Inserting this in (3.4), we get

$$\mathbb{E}\left(e^{\langle T^{-1}Y(T)+m_0\ln(T),\theta\rangle}\right) = e^{\frac{2}{\pi}\int_{\mathbb{S}}\langle\theta,s\rangle\ln|\langle\theta,s\rangle|\rho(ds)}. \tag{3.6}$$

It follows that, for all $s \leq 0$, we have

$$\begin{aligned}\mathbb{E}\left(e^{sT+\langle T^{-1}Y(T)+m_0\ln(T),\theta\rangle}\right) &= \int_0^{+\infty}\mathbb{E}\left(e^{st+\langle t^{-1}Y(t)+m_0\ln(t),\theta\rangle}\Big|T=t\right)f_T(t)dt\\ &= \int_0^{+\infty}e^{st}\mathbb{E}\left(e^{\langle Y(t),t^{-1}\theta\rangle}\right)e^{\ln(t)\langle m_0,\theta\rangle}f_T(t)dt\\ &= \int_0^{+\infty}e^{st}e^{tk_{\mu_1}\left(t^{-1}\theta\right)}e^{\ln(t)\langle m_0,\theta\rangle}f_T(t)dt.\end{aligned}$$

Combined with (3.5), this gives

$$\begin{aligned}\mathbb{E}\left(e^{sT+\langle T^{-1}Y(T)+\ln(T)m_0,\theta\rangle}\right) &= \int_0^{+\infty}e^{st}e^{\frac{2}{\pi}\int_{\mathbb{S}}\langle\theta,s\rangle\ln|\langle\theta,s\rangle|\rho(ds)}f_T(t)dt\\ &= e^{\frac{2}{\pi}\int_{\mathbb{S}}\langle\theta,s\rangle\ln|\langle\theta,s\rangle|\rho(ds)}\mathbb{E}(e^{sT}).\end{aligned}$$

According to (3.6), we get, for all $s \leq 0$,

$$\mathbb{E}\left(e^{sT+\langle T^{-1}Y(T)+\ln(T)m_0,\theta\rangle}\right) = \mathbb{E}\left(e^{\langle T^{-1}Y(T)+\ln(T)m_0,\theta\rangle}\right)\mathbb{E}\left(e^{sT}\right).$$

Consequently, $T^{-1}Y(T)+m_0\ln(T)$ and T are independent.

($\Longleftarrow$) As $T^{-1}Y(T)+m_0\ln(T)$ and T are independent then, for all $s \leq 0$, we have

$$\mathbb{E}\left(e^{sT+\langle T^{-1}Y(T)+m_0\ln(T),\theta\rangle}\right) = \mathbb{E}\left(e^{sT}\right)\mathbb{E}\left(e^{\langle T^{-1}Y(T)+m_0\ln(T),\theta\rangle}\right).$$

Then, we get

$$\int_0^{+\infty}\mathbb{E}\left(e^{st+\langle t^{-1}Y(t)+m_0\ln(t),\theta\rangle}\Big|T=t\right)f_T(t)dt = \mathbb{E}\left(e^{sT}\right)\int_0^{+\infty}\mathbb{E}\left(e^{\langle t^{-1}Y(t)+m_0\ln(t),\theta\rangle}\Big|T=t\right)f_T(t)dt.$$

This implies that

$$\int_0^{+\infty}e^{st+\ln(t)\langle m_0,\theta\rangle}\mathbb{E}\left(e^{\langle Y(t),t^{-1}\theta\rangle}\right)f_T(t)dt = \mathbb{E}\left(e^{sT}\right)\int_0^{+\infty}e^{\ln(t)\langle m_0,\theta\rangle}\mathbb{E}\left(e^{\langle Y(t),t^{-1}\theta\rangle}\right)f_T(t)dt.$$

It follows that

$$\int_0^{+\infty}e^{st+\ln(t)\langle m_0,\theta\rangle}e^{tk_{\mu_1}\left(t^{-1}\theta\right)}f_T(t)dt = \mathbb{E}\left(e^{sT}\right)\int_0^{+\infty}e^{\ln(t)\langle m_0,\theta\rangle}e^{tk_{\mu_1}\left(t^{-1}\theta\right)}f_T(t)dt. \tag{3.7}$$

Now, let's consider the positive measure

$$\mu_{\theta,m_0}(dt) = e^{\ln(t)\langle m_0,\theta\rangle} e^{tk_{\mu_1}\left(t^{-1}\theta\right)} f_T(t) 1_{(0,+\infty)}(t)dt. \tag{3.8}$$

Inserting this in (3.7), we obtain, for all $s \leq 0$,

$$L_{\mu_{\theta,m_0}}(s) = \int_0^{+\infty} e^{st}\mu_{\theta,m_0}(dt) = \mathbb{E}\left(e^{sT}\right)\mu_{\theta,m_0}((0,+\infty)).$$

Consequently, the Laplace transform of the probability measure $\dfrac{\mu_{\theta,m_0}(dt)}{\mu_{\theta,m_0}((0,+\infty))}$ is $\mathbb{E}\left(e^{sT}\right)$. Hence, for all $t > 0$, we have

$$\frac{\mu_{\theta,m_0}(dt)}{\mu_{\theta,m_0}((0,+\infty))} = f_T(t)1_{(0,+\infty)}(t)dt.$$

Then, according to (3.8), we get

$$\frac{e^{\ln(t)\langle m_0,\theta\rangle} e^{tk_{\mu_1}\left(t^{-1}\theta\right)} f_T(t)1_{(0,+\infty)}(t)dt}{\mu_{\theta,m_0}((0,+\infty))} = f_T(t)1_{(0,+\infty)}(t)dt.$$

Using (3.8), we deduce that the function

$$t \longmapsto e^{tk_{\mu_1}\left(t^{-1}\theta\right)+\ln(t)\langle m_0,\theta\rangle}$$

is constant, that is, for all $t > 0$,

$$e^{tk_{\mu_1}\left(t^{-1}\theta\right)+\ln(t)\langle m_0,\theta\rangle} = \mu_{\theta,m_0}((0,+\infty)).$$

Differentiating this with respect to t, we obtain

$$0 = \frac{\partial}{\partial t}\left(tk_{\mu_1}\left(t^{-1}\theta\right) + \ln(t)\langle m_0,\theta\rangle\right) = k_{\mu_1}\left(t^{-1}\theta\right) - \left\langle k'_{\mu_1}\left(t^{-1}\theta\right), t^{-1}\theta\right\rangle + \langle t^{-1}\theta, m_0\rangle.$$

Therefore,

$$k_{\mu_1}\left(t^{-1}\theta\right) = \left\langle k'_{\mu_1}\left(t^{-1}\theta\right), t^{-1}\theta\right\rangle + \langle t^{-1}\theta, m_0\rangle.$$

This with (1.5), implies that $Y = (Y(t))_{t\geq 0}$ is $1-$stable process. □

3.2 Some characteristic properties of a stochastic process on $\mathbb{R}^d$

In this section, we consider a stochastic process defined by $\widetilde{Y} = (\widetilde{Y}(t))_{t\geq 0} = (Y_1(t), \overline{Y}(t))_{t\geq 0}$ on $(0,+\infty)\times\mathbb{R}^{d-1}$ with $d \geq 2$ such that $(Y_1(t))_{t\geq 0}$ and $(\overline{Y}(t)|Y_1(t))_{t\geq 0}$ are Lévy processes. We characterize this process in terms of Laplace transform by using an independence property between a standard exponential random variable T and $\widetilde{Y}(T)$. Moreover, we give some characteristic properties of $\widetilde{Y}(T)$ and $\overline{Y}(T)$. We denote the distribution of $\widetilde{Y}(t)$ by

$$\mu_t(dy_1, d\overline{y}) = \pi_t(dy_1)Q_{y_1,t}(d\overline{y}), \tag{3.9}$$

where $Q_{y_1,t}$ represents the conditional distribution of $(\overline{Y}(t) \mid Y_1(t))$ and π_t is the distribution of $Y_1(t)$. In what follows, we denote by $F_t = F(\mu_t)$ the NEF generated by μ_t and by $(Z(t))_{t\geq 0} = (\overline{Y}(t) \mid Y_1(t))_{t\geq 0}$.

Theorem 3.2.1 *Let $d \geq 2$ and let $(\widetilde{Y}(t))_{t\geq 0} = (Y_1(t), \overline{Y}(t))_{t\geq 0}$ be a stochastic process on $(0,+\infty)\times\mathbb{R}^{d-1}$ pair-wise independent of a standard exponential random variable T. Suppose that the NEF $F_t = F(\mu_t)$ generated by μ_t has a cut on $(0,+\infty)$. Then, for any fixed $p > 1$, $T^{-p}Y_1(T) \perp Y_1(T)$ and $T^{-\frac{1}{2}}Z(T) \perp T$ if, and only if, there exist $\beta > 0$ and $(d-1)$-symmetric positive matrices B_0 and B_1 such that, for all $(\theta_1, \overline{\theta}) \in \{(\theta_1, \overline{\theta}) \in \mathbb{R}^d \ ; \ \theta_1 + \frac{t}{2}\overline{\theta}^* B_1\overline{\theta} < 1/\beta\}$*

$$k_{\mu_t}(\theta_1, \overline{\theta}) = t\left(1 - \left(1 - \beta\left(\theta_1 + \frac{t}{2}\overline{\theta}^* B_1\overline{\theta}\right)\right)^{\frac{1}{p}} + \frac{1}{2}\overline{\theta}^* B_0\overline{\theta}\right),$$

where $\overline{\theta}^*$ denotes the transpose of the vector $\overline{\theta}$.

Before proving this theorem, we need the following intermediary technical result.

Lemma 3.2.2 *Let $d \geq 2$ and let $(\widetilde{Y}(t))_{t\geq 0} = (Y_1(t), \overline{Y}(t))_{t\geq 0}$ be a stochastic process on $(0,+\infty)\times\mathbb{R}^{d-1}$ pair-wise independent of a standard exponential random variable T. Suppose that $F_t = F(\mu_t)$ has a cut on $(0,+\infty)$ and that $T^{-\frac{1}{2}}Z(T) \perp T$. Then, there exist $(d-1)$-symmetric positive matrices B_0 and B_1, such that the conditional distribution $Q_{y_1,t}$ is centered normal with a variance-covariance matrix $t(B_0 + y_1B_1)$.*

Proof of Lemma 3.2.2

Since $(Y_1(t), \overline{Y}(t))_{t\geq 0}$ is pair-wise independent of T then, the random variables $Z(t)$ and T are independent. By hypothesis, we have $T^{-\frac{1}{2}}Z(T) \perp T$. Therefore, using Theorem 3.1.1,

we deduce that, $(Z(t))_{t\geq 0}$ is a strictly 2-stable process on $\mathbb{R}^{d-1}$, this means that there exists a variance-covariance matrix $\Sigma(y_1)$ depending on y_1, such that the conditional distribution of $\overline{Y}(t)$, given $Y_1(t) = y_1$ is centered normal, $N(0, t\Sigma(y_1))$, with a variance-covariance matrix $t\Sigma(y_1)$.

Taking $t = 1$, the cumulant function of $Q_{y_1,1}$ is

$$k_{Q_{y_1,1}}(\overline{\theta}) = \frac{1}{2}\overline{\theta}^*\Sigma(y_1)\overline{\theta}, \quad \text{for all} \quad \overline{\theta} \in \mathbb{R}^{d-1}. \tag{3.10}$$

Since F_t has a cut on $(0, +\infty)$, then using (1.2), we deduce that there exist H and K, such that the conditional cumulant function of $Q_{y_1,1}$ is given by

$$k_{Q_{y_1,1}}(\overline{\theta}) = K(\overline{\theta}) - y_1 H(\overline{\theta}).$$

Combined with (3.10), this gives

$$\frac{1}{2}\overline{\theta}^*\Sigma(y_1)\overline{\theta} = K(\overline{\theta}) - y_1 H(\overline{\theta}).$$

Applying the differential twice with respect to $\overline{\theta}$, we get for all $\overline{\theta} \in \mathbb{R}^{d-1}$ and for all $y_1 > 0$,

$$\Sigma(y_1) = K''(\overline{\theta}) - y_1 H''(\overline{\theta}). \tag{3.11}$$

Then, the function $y_1 \longrightarrow \Sigma(y_1)$ is affine and independent of $\overline{\theta}$. Since $\Sigma(y_1)$ is a $(d-1)-$symmetric definite positive matrix, then, letting $y_1 \to 0^+$, we deduce that $K''(\overline{\theta})$ has also the same characteristic property, *i.e.* being a $(d-1)-$symmetric positive matrix independent of $\overline{\theta}$. This implies the existence of a matrix B_0 ensuring the same characteristic property, such that $K''(\overline{\theta}) = B_0$. Combining this with (3.11) leads to the existence of another matrix B_1 independent of $\overline{\theta}$ such that $B_1 = -H''(\overline{\theta})$. Furthermore, for all $h \in \mathbb{R}^{d-1}$,

$$h^*\Sigma(y_1)h = h^*B_0h + y_1h^*B_1h \geq 0.$$

Otherwise, we get for all $y_1 \in (0, +\infty)$,

$$\frac{h^*B_0h}{y_1} \geq -h^*B_1h.$$

Hence, letting $y_1 \to +\infty$, we have $h^* B_1 h \geq 0$. This implies that B_1 is also a symmetric positive matrix. □

Now, we are ready to prove Theorem 3.2.1.

Proof of Theorem 3.2.1

($\Longrightarrow$) Since for any fixed $p > 1$, $T^{-p} Y_1(T)$ and $Y_1(T)$ are independent then, according to Letac and Seshadri (2005), there exists $\beta > 0$ such that, for all $s < \frac{1}{\beta}$,

$$L_{\pi_t}(s) = \mathbb{E}(e^{sY_1(t)}) = e^{t(1-(1-\beta s)^{\frac{1}{p}})}. \tag{3.12}$$

Moreover, using Lemma 3.2.2, we deduce that the conditional distribution $Q_{y_1,t}$ is centered Gaussian with a variance-covariance matrix $t(B_0 + y_1 B_1)$. This implies that, for all $\theta = (\theta_1, \overline{\theta}) \in \Theta(\mu_t) \subset \mathbb{R} \times \mathbb{R}^{d-1}$, we have

$$\begin{aligned}
\mathbb{E}(e^{\langle \theta, \widetilde{Y}(t) \rangle}) &= \mathbb{E}(e^{\theta_1 Y_1(t) + \langle \overline{\theta}, \overline{Y}(t) \rangle}) \\
&= \int_0^{+\infty} \int_{\mathbb{R}^{d-1}} e^{\theta_1 y_1} e^{\langle \overline{\theta}, \overline{y} \rangle} \pi_t(dy_1) Q_{y_1,t}(d\overline{y}) \\
&= \int_0^{+\infty} e^{\theta_1 y_1} \left(\int_{\mathbb{R}^{d-1}} e^{\langle \overline{\theta}, \overline{y} \rangle} Q_{y_1,t}(d\overline{y}) \right) \pi_t(dy_1) \\
&= \int_0^{+\infty} e^{\theta_1 y_1} e^{\frac{t}{2} \overline{\theta}^* (B_0 + y_1 B_1) \overline{\theta}} \pi_t(dy_1) \\
&= e^{\frac{t}{2} \overline{\theta}^* B_0 \overline{\theta}} \int_0^{+\infty} e^{y_1 (\theta_1 + \frac{t}{2} \overline{\theta}^* B_1 \overline{\theta})} \pi_t(dy_1) = e^{\frac{t}{2} \overline{\theta}^* B_0 \overline{\theta}} L_{\pi_t} \left(\theta_1 + \frac{t}{2} \overline{\theta}^* B_1 \overline{\theta} \right).
\end{aligned}$$

Using the Laplace transform of $Y_1(t)$ in (3.12) we get, for all $(\theta_1, \overline{\theta}) \in \Theta(\mu_t) = \{(\theta_1, \overline{\theta}) \in \mathbb{R}^d \ ; \ \theta_1 + \frac{t}{2} \overline{\theta}^* B_1 \overline{\theta} < 1/\beta\}$,

$$L_{\mu_t}(\theta_1, \overline{\theta}) = e^{t \left(1 - \left(1 - \beta \left(\theta_1 + \frac{t}{2} \overline{\theta}^* B_1 \overline{\theta} \right) \right)^{\frac{1}{p}} + \frac{1}{2} \overline{\theta}^* B_0 \overline{\theta} \right)},$$

which is the desired result.

($\Longleftarrow$) By hypothesis, for all $(\theta_1, \overline{\theta}) \in \Theta(\mu_t)$, we have

$$L_{\mu_t}(\theta_1, \overline{\theta}) = e^{t \left(1 - \left(1 - \beta \left(\theta_1 + \frac{t}{2} \overline{\theta}^* B_1 \overline{\theta} \right) \right)^{\frac{1}{p}} + \frac{1}{2} \overline{\theta}^* B_0 \overline{\theta} \right)}. \tag{3.13}$$

Taking $\overline{\theta} = 0$, we get, for all $\theta_1 < 1/\beta$,

$$L_{\pi_t}(\theta_1) = \mathbb{E}(e^{\theta_1 Y_1(t)}) = e^{t(1-(1-\beta\theta_1))^{\frac{1}{p}})}. \tag{3.14}$$

According to Letac and Seshadri (2005), we deduce that, for any fixed $p > 1$, $T^{-p}Y_1(T)$ and $Y_1(T)$ are independent.

Furthermore, on the one hand, from (3.13) and (3.14), we get

$$L_{\mu_t}(\theta_1, \overline{\theta}) = e^{\frac{t}{2}\overline{\theta}^* B_0 \overline{\theta}} L_{\pi_t}\left(\theta_1 + \frac{t}{2}\overline{\theta}^* B_1 \overline{\theta}\right) = \int_0^{+\infty} e^{\theta_1 y_1} e^{\frac{t}{2}\overline{\theta}^*(B_0 + y_1 B_1)\overline{\theta}} \pi_t(dy_1).$$

On the other hand, using (3.9), we deduce that

$$L_{\mu_t}(\theta_1, \overline{\theta}) = \int_0^{+\infty} e^{\theta_1 y_1} \left(\int_{\mathbb{R}^{d-1}} e^{\langle \overline{\theta}, \overline{y}\rangle} Q_{y_1,t}(d\overline{y})\right) \pi_t(dy_1) = \int_0^{+\infty} e^{\theta_1 y_1} L_{Q_{y_1,t}}(\overline{\theta}) \pi_t(dy_1).$$

Then, the Laplace transform of the conditional distribution $Q_{y_1,t}$ is evaluated, for all $\overline{\theta} \in \mathbb{R}^{d-1}$, by

$$L_{Q_{y_1,t}}(\overline{\theta}) = e^{\frac{t}{2}\overline{\theta}^*(B_0 + y_1 B_1)\overline{\theta}}.$$

This means that the conditional distribution of $(Z(t))_{t\geq 0}$ is Gaussian on $\mathbb{R}^{d-1}$. Using Theorem 3.1.1, we deduce that $T^{-\frac{1}{2}} Z(T) \perp T$. □

Remark 3.2.3 *Note that the conditional distribution $Q_{y_1,t}$ of $Z(t)$ is centered normal $N(0, t(B_0 + y_1 B_1))$.*

The next example illustrates the characterization given in Theorem 3.2.1. Let $(\widetilde{Y}(t))_{t\geq 0} = (Y_1(t), \overline{Y}(t))_{t\geq 0}$ be a stochastic process such that $Y_1(t)$ has the inverse Gaussian distribution.

Example 3.2.4 *$(\widetilde{Y}(t))_{t\geq 0} = (Y_1(t), \overline{Y}(t))_{t\geq 0}$ be a stochastic process on $(0, +\infty) \times \mathbb{R}$ such that $Y_1(t)$ is inverse Gaussian with a density function as follows*

$$IG\left(\beta t/2, \beta t^2/2\right)(y_1) = \frac{t\sqrt{\beta}}{2\sqrt{\pi}} y_1^{-\frac{3}{2}} e^{t - \frac{y_1}{\beta} - \frac{\beta t^2}{4y_1}} 1_{(0,+\infty)}(y_1)$$

and $(\overline{Y}(t)|Y_1(t) = y_1) \sim N(0, ty_1)$. Then, the joint density function of $\widetilde{Y}(t)$ is

$$\frac{\sqrt{t\beta}}{2^{\frac{3}{2}}\pi y_1^2} e^{t - \frac{y_1}{\beta} - \frac{\beta t^2}{4y_1} - \frac{\overline{y}^2}{2ty_1}} 1_{(0,+\infty)\times\mathbb{R}}(y_1, \overline{y}).$$

Since, for all $\theta_1 < \frac{1}{\beta}$ and $\overline{\theta} \in \mathbb{R}$, we get

$$L_{Y_1(t)}(\theta_1) = e^{t(1-\sqrt{1-\beta\theta_1})} \quad \text{and} \quad L_{(\overline{Y}(t)|Y_1(t)=y_1)}(\overline{\theta}) = e^{\frac{ty_1\overline{\theta}^2}{2}},$$

it follows that, for all $\theta \in \Theta(\mu_t) = \{\theta = (\theta_1, \overline{\theta}) \in \mathbb{R}^2;\ \theta_1 + t\frac{\overline{\theta}^2}{2} < \frac{1}{\beta}\}$, we have

$$L_{\mu_t}(\theta_1, \overline{\theta}) = e^{t\left(1-\sqrt{1-\beta\left(\theta_1+t\frac{\overline{\theta}^2}{2}\right)}\right)}. \tag{3.15}$$

Conversely, let us consider a stochastic process $(\widetilde{Y}(t))_{t\geq 0}$ on $(0,+\infty)\times\mathbb{R}$ such that (3.15) holds. Using Theorem 3.2.1, we deduce that $Y_1(t)$ is inverse Gaussian distributed with parameters $\beta t/2$ and $\beta t^2/2$ and that $(\overline{Y}(t)|Y_1(t) = y_1)$ is Gaussian distributed with a variance ty_1. It follows that $T^{-2}Y_1(T) \perp Y_1(T)$ and $T^{-\frac{1}{2}}Z(T) \perp T$.

Note that, for $t = 1$, $\overline{Y}(t)$ is normal inverse Gaussian distributed. This class of distributions has been studied by Barndorff-Nielsen (1997).

Next, we prove that the conditional distribution of $\overline{Y}(T)$, given $Y_1(T) = y_1$ is multivariate Laplace distributed. For this purpose, we recall some facts concerning this distribution.

Theorem 3.2.5 *Let $d \geq 2$ and let $(\widetilde{Y}(t))_{t\geq 0} = (Y_1(t), \overline{Y}(t))_{t\geq 0}$ be a stochastic process on $(0,+\infty)\times\mathbb{R}^{d-1}$ pair-wise independent of a standard exponential random variable T. Suppose that $F_t = F(\mu_t)$ has a cut on $(0,+\infty)$ and that $T^{-\frac{1}{2}}Z(T) \perp T$. Then the conditional random vector $Z(T)$ is multivariate Laplace distributed with a location parameter $\sigma = 0$ and a variance-covariance parameter $B_0 + y_1B_1$, where B_0 and B_1 are $(d-1)$-symmetric positive matrices.*

Proof

Using Lemma 3.2.2, we deduce that $Z(t)$ is centered Gaussian with the following variance-covariance matrix $t(B_0 + y_1B_1)$, and since T is a standard exponential random variable, then we have

$$\mathbb{E}\left(e^{\langle\overline{\theta},\overline{Y}(T)\rangle}\Big|Y_1(T) = y_1\right) = \int_0^{+\infty} e^{-t}\mathbb{E}\left(e^{\langle\overline{\theta},\overline{Y}(t)\rangle}\Big|Y_1(t) = y_1\right)dt = \int_0^{+\infty} e^{-t}e^{\frac{t}{2}\overline{\theta}^*(B_0+y_1B_1)\overline{\theta}}dt.$$

It follows that, for all $\overline{\theta} \in \mathbb{R}^{d-1}$, we have

$$\mathbb{E}\left(e^{\langle\overline{\theta},\overline{Y}(T)\rangle}\Big|Y_1(T) = y_1\right) = \frac{1}{1-\frac{1}{2}\overline{\theta}^*(B_0+y_1B_1)\overline{\theta}}.$$

Combined with (1.9), this implies that $Z(T)$ is multivariate Laplace distributed with a location parameter $\sigma = 0$ and a variance-covariance parameter $B_0 + y_1 B_1$. □

Next, we give the density function of $\widetilde{Y}(T)$. In what follows, we denote the density function of $\widetilde{Y}(T)$ by $f_{\widetilde{Y}(T)}(y_1, \overline{y})$, and the density function of $\overline{Y}(T)$ by $f_{\overline{Y}(T)}(\overline{y})$.

Corollary 3.2.6 *Let $d \geq 2$ and let $(\widetilde{Y}(t))_{t\geq 0} = (Y_1(t), \overline{Y}(t))_{t\geq 0}$ be a stochastic process on $(0, +\infty) \times \mathbb{R}^{d-1}$ pair-wise independent of a standard exponential random variable T. Suppose that $F_t = F(\mu_t)$ has a cut on $(0, +\infty)$, $T^{-\frac{1}{2}} Z(T) \perp T$ and, for any fixed $p > 1$, $T^{-p} Y_1(T) \perp Y_1(T)$. Then, the density function of $\widetilde{Y}(T)$ is given by*

$$f_{\widetilde{Y}(T)}(y_1, \overline{y}) = \frac{1}{(2\pi)^{\frac{d-1}{2}}} \frac{2K_{\frac{d-3}{2}}\left(\sqrt{2\overline{y}^*(B_0 + y_1 B_1)^{-1}\overline{y}}\right)}{\left(\sqrt{\frac{1}{2}\overline{y}^*(B_0 + y_1 B_1)^{-1}\overline{y}}\right)^{\frac{d-3}{2}}} \frac{y_1^{\frac{1}{p}-1} e^{-\frac{y_1}{\beta}}}{\beta^{\frac{1}{p}}\Gamma(\frac{1}{p})},$$

where $\beta > 0$, $K_{\frac{d-3}{2}}$ denotes the modified Bessel function of the third kind with order $\frac{d-3}{2}$, and where B_0 and B_1 are $(d-1)$-symmetric positive matrices.

Proof

Using Theorem 3.2.5, we deduce that the random vector $\left(\overline{Y}(T)\middle| Y_1(T)\right)$ is multivariate Laplace distributed, with a density function

$$\frac{1}{(2\pi)^{\frac{d-1}{2}}} \frac{2K_{\frac{d-3}{2}}\left(\sqrt{2\overline{y}^*(B_0 + y_1 B_1)^{-1}\overline{y}}\right)}{\left(\sqrt{\frac{1}{2}\overline{y}^*(B_0 + y_1 B_1)^{-1}\overline{y}}\right)^{\frac{d-3}{2}}}.$$

According to Letac and Seshadri (2005), $Y_1(T)$ is Gamma distributed, with a scale parameter β and a shape parameter $1/p$. Therefore, the joint density function of $\widetilde{Y}$ is given by

$$\begin{aligned} f_{\widetilde{Y}(T)}(y_1, \overline{y}) &= f_{(Y_1(T), \overline{Y}(T))}(y_1, \overline{y}) \\ &= \frac{1}{(2\pi)^{\frac{d-1}{2}}} \frac{2K_{\frac{d-3}{2}}\left(\sqrt{2\overline{y}^*(B_0 + y_1 B_1)^{-1}\overline{y}}\right)}{\left(\sqrt{\frac{1}{2}\overline{y}^*(B_0 + y_1 B_1)^{-1}\overline{y}}\right)^{\frac{d-3}{2}}} \frac{y_1^{\frac{1}{p}-1} e^{-\frac{y_1}{\beta}}}{\beta^{\frac{1}{p}}\Gamma(\frac{1}{p})}. \end{aligned}$$

□

Remark 3.2.7 *Using the density function of $\widetilde{Y}(T)$, we deduce that the marginal density function of $\overline{Y}(T)$ is given by*

$$f_{\overline{Y}(T)}(\overline{y}) = \int_0^{+\infty} \frac{1}{(2\pi)^{\frac{d-1}{2}}} \frac{2K_{\frac{d-1}{2}-1}\left(\sqrt{2\overline{y}'(B_0+y_1B_1)^{-1}\overline{y}}\right)}{\left(\sqrt{\frac{1}{2}\overline{y}'(B_0+y_1B_1)^{-1}\overline{y}}\right)^{\frac{d-1}{2}-1}} \frac{y_1^{\frac{1}{p}-1}e^{-\frac{y_1}{\beta}}}{\beta^{\frac{1}{p}}\Gamma(\frac{1}{p})}dy_1.$$

Finally, we give an example which illustrates the result of the previous corollary.

Example 3.2.8 *Let $(\widetilde{Y}(t))_{t\geq 0} = (Y_1(t), \overline{Y}(t))_{t\geq 0}$ be a stochastic process on $(0,+\infty)\times\mathbb{R}$ pair-wise independent of a standard exponential random variable T. Suppose that $Y_1(T)$ is Gamma distributed and that $(\overline{Y}(T)|Y_1(T) = y_1)$ has the Laplace distribution on $\mathbb{R}$ defined by $\frac{e^{-\frac{|\overline{y}|}{y_1}}}{2y_1}d\overline{y}$. Then, applying the previous corollary, we conclude that the density function of $\widetilde{Y}(T) = (Y_1(T), \overline{Y}(T))$ is given by*

$$f_{\widetilde{Y}(T)}(y_1,\overline{y}) = f_{(Y_1(T),\overline{Y}(T))}(y_1,\overline{y}) = \frac{e^{-\frac{|\overline{y}|}{y_1}}}{2y_1}\frac{y_1^{\frac{1}{p}-1}e^{-\frac{y_1}{\beta}}}{\beta^{\frac{1}{p}}\Gamma(\frac{1}{p})}1_{(0,+\infty)\times\mathbb{R}}(y_1,\overline{y})d\overline{y}dy_1.$$

Consequently, the marginal density function of $\overline{Y}(T)$ is

$$f_{\overline{Y}(T)}(\overline{y}) = \int_0^{+\infty} \frac{e^{-\frac{|\overline{y}|}{y_1}}}{2y_1}\frac{y_1^{\frac{1}{p}-1}e^{-\frac{y_1}{\beta}}}{\beta^{\frac{1}{p}}\Gamma(\frac{1}{p})}dy_1 = \frac{\Gamma(1-\frac{1}{p})}{2\beta^{\frac{1}{p}}\Gamma(\frac{1}{p})|\overline{y}|^{1-\frac{1}{p}}}\int_0^{+\infty} e^{-\frac{y_1}{\beta}}\frac{|\overline{y}|^{1-\frac{1}{p}}e^{-\frac{|\overline{y}|}{y_1}}}{\Gamma(1-\frac{1}{p})y_1^{2-\frac{1}{p}}}dy_1.$$

Using the expression of the Laplace transform of the Inverse Gamma distribution, we deduce that

$$f_{\overline{Y}(T)}(\overline{y}) = \frac{\Gamma(1-\frac{1}{p})}{2\beta^{\frac{1}{p}}\Gamma(\frac{1}{p})|\overline{y}|^{1-\frac{1}{p}}}L_{I\gamma(1-\frac{1}{p},|\overline{y}|)}(-\frac{1}{\beta}) = \frac{|\overline{y}|^{\frac{1}{2}(\frac{1}{p}-1)}}{\sqrt{\beta}\Gamma(\frac{1}{p})}K_{1-\frac{1}{p}}\left(\sqrt{\frac{4|\overline{y}|}{\beta}}\right),$$

where $K_{1-\frac{1}{p}}$ denotes the modified Bessel function of the third kind with order $(1-\frac{1}{p})$ (for more details, see Seshadri (1994), p. 27).

.

Conclusion and perspectives

In this thesis, we have introduced a family of distributions called the normal α-stable family. It is the normal variance-mean mixtures where the mixing densities are the drifted α-stable distributions. It is important to notice that this family extends the normal inverse Gaussian one ($\alpha = 1/2$) introduced by Barndorff-Nielsen (1997). Some important characteristic properties of this extended class of distributions are given. More precisely, we have characterized the natural exponential family (NEF) generated by the normal α-stable distribution by its variance function which is explicitly written as a function of the mean. Furthermore, since the normal α-stable distribution is infinitely divisible, then using the link between this property and the Lévy measure, we have given the Lévy measure associated to the normal α-stable distribution using the Lévy-Khintchine representation. Moreover, we have approximated the normal $\alpha-$stable density using the Monte-Carlo method. We have described various methods of estimating parameters, such as the moments and the tailed regression methods. The asymptotic results of the obtained estimators also supported by simulation studies. In addition, we have illustrated our estimation by the histograms of the different estimators $\widehat{\alpha}$ and $\widehat{\beta}$ of α and β. Moreover, we have computed the mean squared error and the variation coefficient for each method. Hence, we have given the curves of the MSE of $\widehat{\alpha}$ and $\widehat{\beta}$ and the curves of the estimate and true densities for different values of α and β. In this work, we have considered the problem of parameters' estimations for the normal α-stable model using the moments and tailed regression methods. In the future, we will investigate to estimate the parameters by applying the maximum likelihood approach of the EM type. This needs also to be developed

by further work with other estimation methods. It is possible to construct a number of different estimators under different assumptions and introduce the Shrinkage estimators. We will apply our model to finance domain, mainly to study the stochastic prices volatility.

In terms of processes, we have given a characterization of a multivariate stochastic stable process $(Y(t))_{t\geq 0}$ using an independence property between a positive random variable T and $Y(T)$. In fact, we characterize the extension of this process in the multivariate case using the stable distributions characterization given by Hassairi and Louati (2009). In fact, we characterize the class of stable processes in a multidimensional setting as those processes $Y = (Y(t))_{t\geq 0}$ that guarantee the independence of $T^{-1/\alpha}Y(T) + A_\alpha(T)$ and T with $\alpha \in (0, 2]$, where T is a positive random variable independent of $Y = (Y(t))_{t\geq 0}$ and $A_\alpha(t) = 0$ unless $\alpha = 1$, in which case it is proportional to $\log(t)$. This extension covers the whole possible range of α, allowing a multivariate and not requiring Y components to be positive. Then, we establish, in terms of Laplace transform, a characterization of a multivariate stochastic process $(\widetilde{Y}(t))_{t\geq 0} = (Y_1(t), \overline{Y}(t))_{t\geq 0}$ on $(0, +\infty) \times \mathbb{R}^{d-1}$, with $d \geq 2$, based on the cut property. Moreover, we give some properties of the distribution $\overline{Y}(t)$ which is defined as a normal variance-mean mixture with a stable mixing distribution and we determine the distribution of $\widetilde{Y}(T)$. These results extend the one dimensional case given by Letac and Seshadri (2005).

Bibliography

Araujo, A. and Giné, E. (1980). *The Central Limit Theorem for Real and Banach Valued Random Variables*. New York. Wiley.

Barndorff-Nielsen, O.E. and Koudou, A.E. (1995). Cuts in natural exponential families. *Theory. Probab. Appl.* **40**, 361-372.

Barndorff-Nielsen, O.E. (1997). Normal inverse Gaussian distributions and stochastic volatility modeling. *Scan. J. Stat.* **24**, 1-13.

Barndorff-Nielsen, O.E. and Shephard, N. (2001). *Normal Modified Stable Processes*. Department of Economics. Oxford University.

Bertoin, J. (1996). *Lévy Processes*. Cambridge University Press. Cambridge.

Brown, L.D. (1986). *Fundamentals of statistical exponential families with applications in statistical decision theory*. IMS. Lecture Notes and Monographs Series, Hayward.

Casalis, M. (1990). The $(2d + 4)$ simple quadratic natural exponential families on $\mathbb{R}^d$. *Ann. Statist.* **24**, 1828-1854.

Christoph, G. and Wolf, W. (1992). *Convergence theorems with stable limit law*. Berlin. Akadamie Verlag.

Embrechts, P., Klüppelberg, C. and Mikosch, T. (1997). *Modelling extreme events for insurance and finance*. Springer-Verlag. Berline.

Fama, E. (1965). The behavior of stock market prices. *J. Business.* **38**, 34-105.

Fama, E. and Roll, R. (1971). Parameter Estimates for Symmetric Stable Distributions. *Journal of the American Statistical Association***66**, 331-338.

Feller, W. (1971) *An introduction to probability theory and its applications.* Tome 2. Wiley. New York.

Forsberg, L. and Bollerslev, T. (2002). Bridging the gap between the distribution of realized (ECU) volatility and ARCH modelling (of the EURO): The GARCH-NIG model. *J. Appl. Econ.* **17**, 535-548.

Hassairi, A. (1992). La classification des familles exponentielles naturelles sur $\mathbb{R}^n$ par l'action du groupe linéaire de $\mathbb{R}^{n+1}$. *Comptes Rendus de l'Académie des Sciences. Paris, t. 315, série I.* 207-210.

Hassairi, A. (1999). Generalized variance and exponential families. *Ann. Statist.* **27**, 374-385.

Hassairi, A. and Zarai, M. (2004). Characterization of the cubic exponential families by orthogonality of polynomials. *Ann. Probab.* **32(3B)**, 2463-2476.

Hassairi, A. and Louati, M. (2009). Multivariate stable exponential families and Tweedie scale. *J. Stat. Plann. Inf.* **139(2)**, 143-158.

Hougaard, P. (1986). Survival models for homogeneous populations derived from stable distributions. *Biometrika.* **73**, 387-396.

Johnson, N.L., Kotz S. and Balakrishnan, N. (1994) *Continuous univariate distributions.* Wiley. New York.

Kokonendji, C.C. and Masmoudi, A. (2006). A Characterization of Poisson-Gaussian families by generalized variance. *Bernoulli.* **12**, 371-379.

Kokonendji, C.C. and Masmoudi. A. (2013). On the Monge-Ampère equation for characterizing gamma-Gaussian model. *Statist. Prob. Letters.* **83**, 1692-1698.

Letac, G. (1989). Le problème de la classification des familles exponentielles naturelles sur $\mathbb{R}^d$ ayant une fonction variance quadratique. *Probability Measures on Groups IX. Lecture Notes in Math.* **1306**, 194-215. Springer, Berline.

Letac, G. and Mora, M. (1990). Natural real exponential families with cubic variance function. *Ann. Statist.* **18**, 1-37.

Letac, G.,(1992). *Lecture on natural exponential families and their variance functions.* Monografias de Mathematica 50. IMPA, Rio de Janeiro.

Letac, G. and Seshadri, V. (2005). Exponential stopping and drifted stable processes. *Statist. Probab. Lett.* **72**, 137-143.

Louati, M. (2013). Mixture and reciprocity of exponential models. *Statist. Probab. Lett.* **83(2)**. 452-458.

Louati, M., Masmoudi, A. and Mselmi, F. (2015). Multivariate normal $\alpha-$stable exponential families. *Mediterr. J. Math.* (DOI) 10.1007/s00009-015-0562-y.

McCulloch, J.H. (1996). 13 Financial applications of stable distributions. *Handbook of Statistics.* **14**, 393-425.

McCulloch, J.H., (1986). Simple consistent estimators of stable distribution parameters. *Comm. in Statistic-Simulation and Computation* **4**, 1109-1138.

Morris, C.N. (1982). Natural exponential families with quadratic variance functions. *Ann. Statist.* **10**, 65-80.

Nolan, J. (2003). *Modeling financial distributions with stable distributions.* Volume 1 of Handbooks in Finance, Chapter 3, pp. 105-130. Amsterdam. Elsevier.

Nolan, J. (2007). *Stable distributions: Models for heavy tailed data.* Brikhäuser.

Pearson, K. (1936). Methods of moments and method of maximum likelihood. *Biometrika.* **28**, 34-59.

Press, J.S. (1972). Estimation in Univariate and Multivariate Stable Distributions. *Journal of the American Statistical Association.* **67** 842-846.

Samorodnitsky, G. and Taqqu, M.S. (1994). *Stable Non-Gaussian Random Processes: Stochastic Models with Infinite Variance.* Chapman and Hall. New York, London.

Sato, K.I. (1999). *Lévy Processes and infinitely divisible distributions.* Cambridge University Press. Cambridge.

Seshadri, V. (1994). *The Inverse Gaussian Distribution. A case study in exponential families.* Oxford University Press. New York.

Shiryayev, A.N. (1984). *Probability.* volume 95 of Graduate Texts in Mathematics. Springer-Verlag.

Torbjørn, E. Taesu, K. and Te-Won, L. (2006). On the Multivariate Laplace Distribution. *IEEE signal processing letters.* **13**, 300-303.

Tweedie, M.C.K. (1947). Functions of a statistical variate with given means with special reference to Laplacian distribution. *Proc. Cambridge Philos. Soc.* **43**, 41-49.

Wilhelmsson, A. (2009). Value at Risk with time varying variance, skewness and kurtosis the NIG-ACD model. *Econometrics Journal.* **12(1)**, 82-104.

Zolotarev, V.M. (1986). *One-dimensional Stable Distributions.* Volume 65 of Translations of mathematical monographs. American Mathematical Society. Translation from the original 1983 Russian edition.

Zolotarev, V.M. (1995). On representation of densities of stable laws by special functions. *Theory Probab. Appl.* **39**, 354-362.